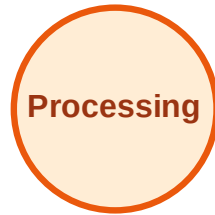


BFS Traversal

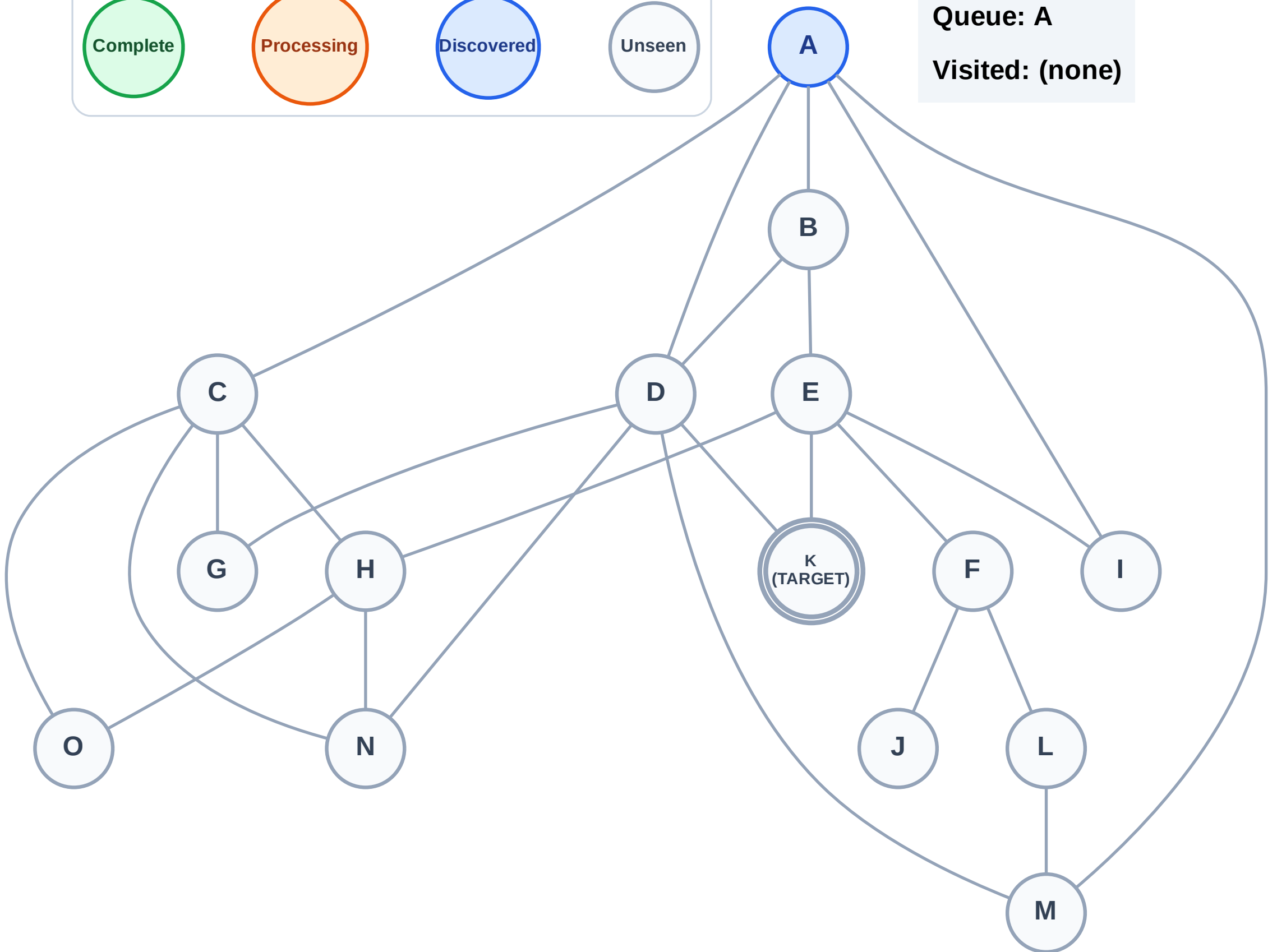
Step 1: Start at A (Seeking target node K)

Legend

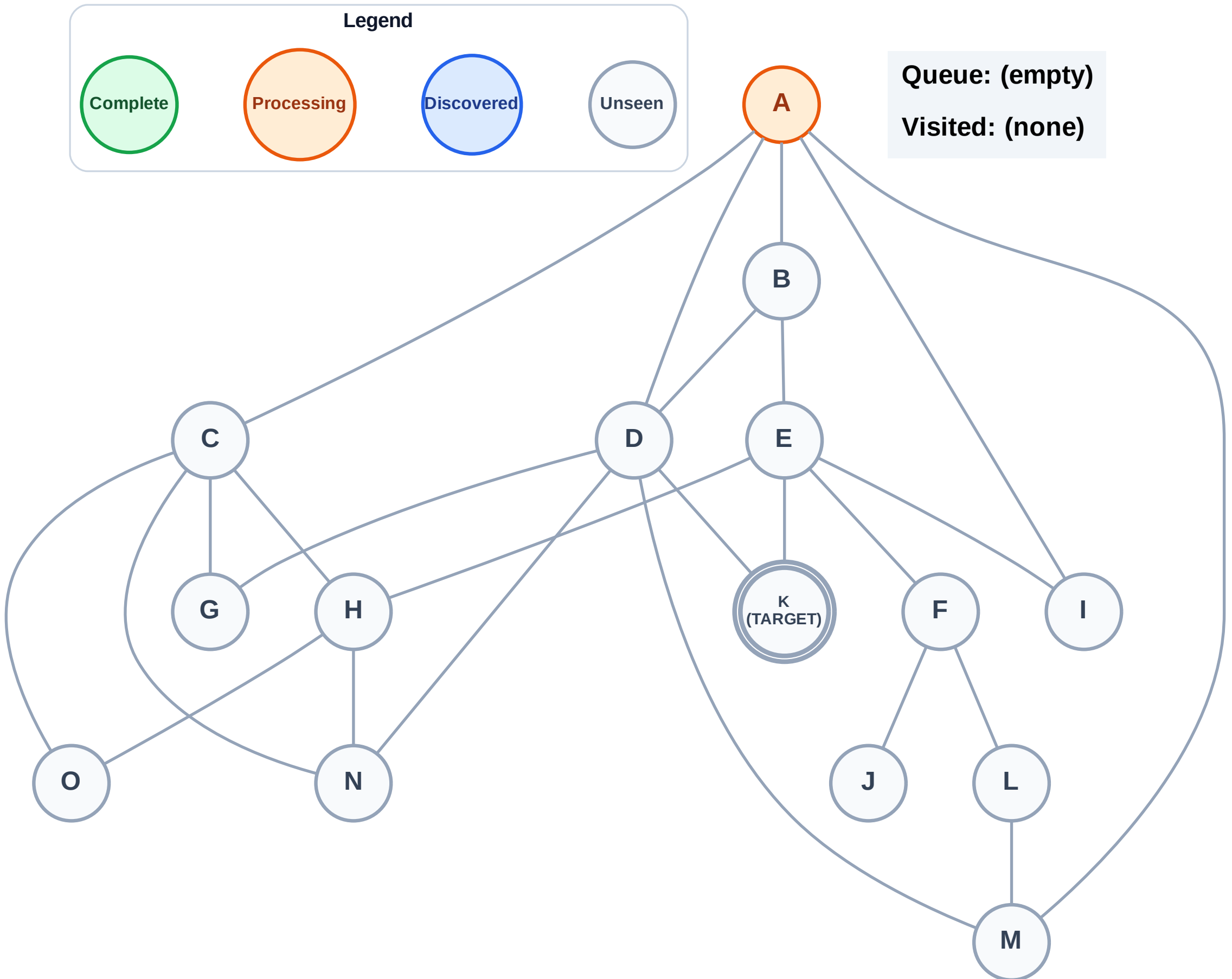


Queue: A

Visited: (none)



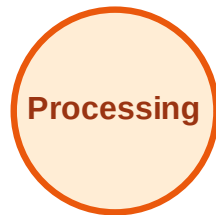
Step 2: Process node A



BFS Traversal

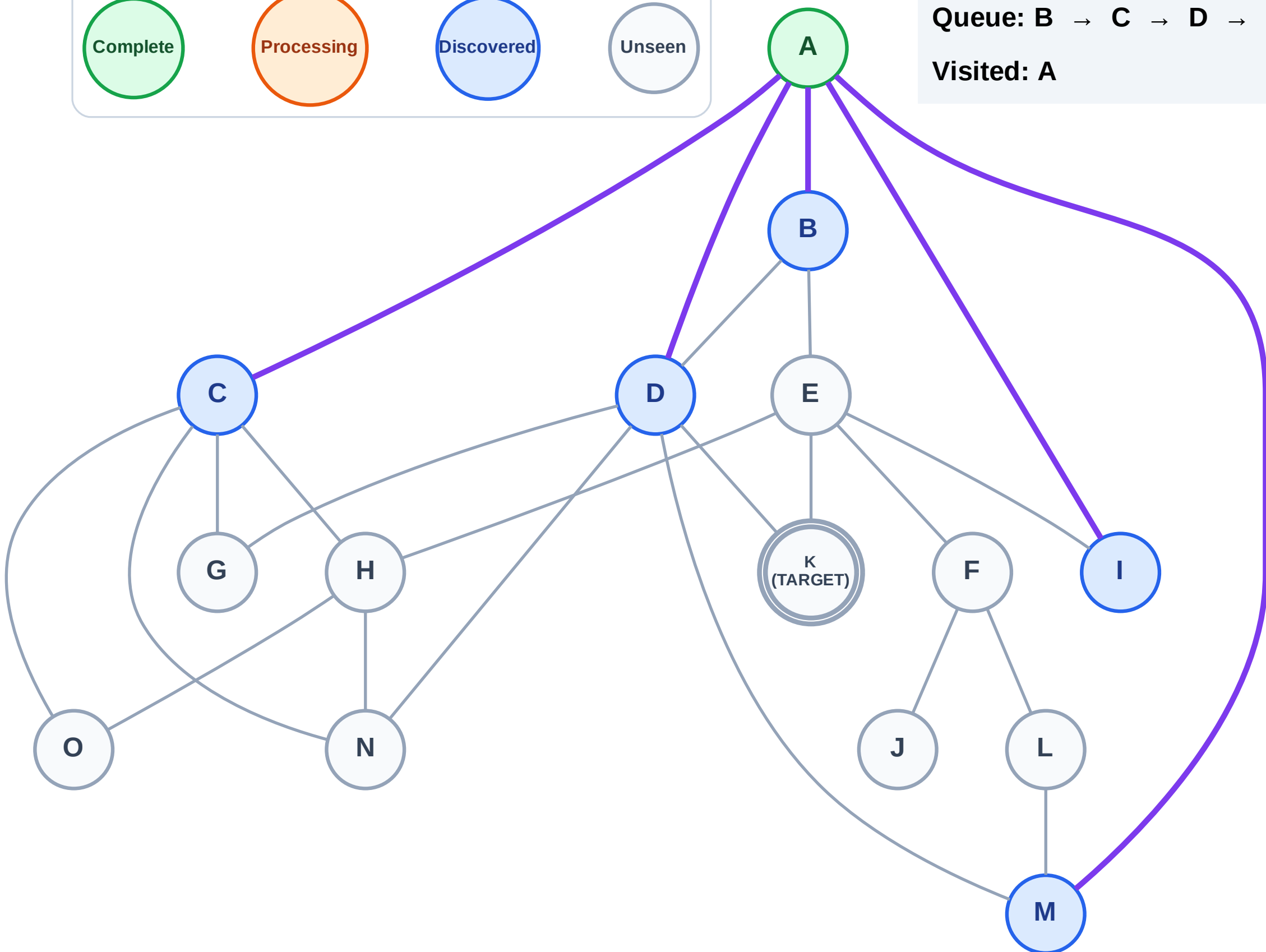
Step 3: Discover B, C, D, I, M from A

Legend



Queue: B → C → D → I → M

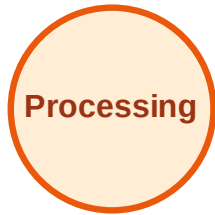
Visited: A



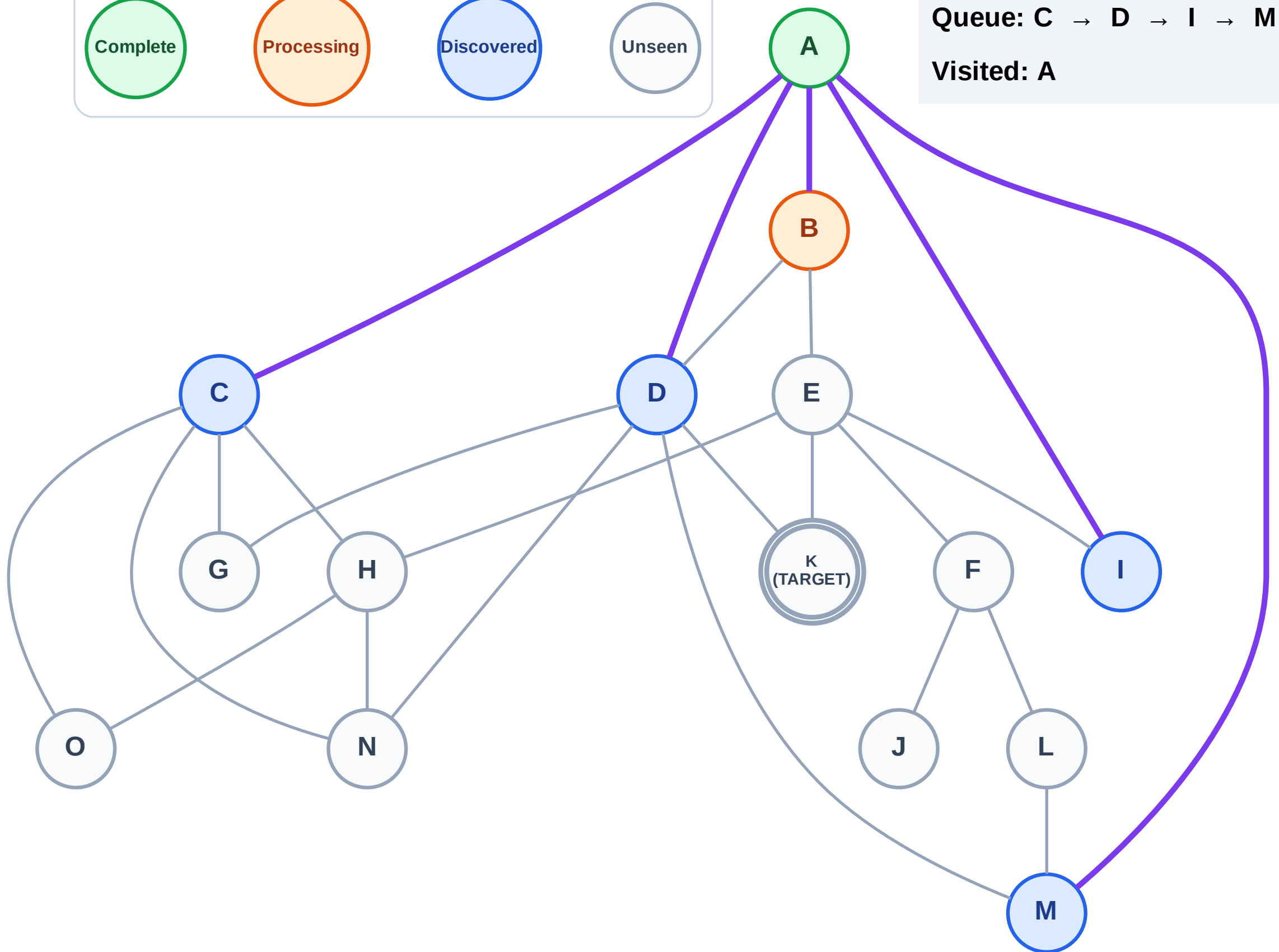
BFS Traversal

Step 4: Process node B

Legend



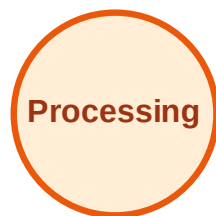
Queue: C → D → I → M
Visited: A



BFS Traversal

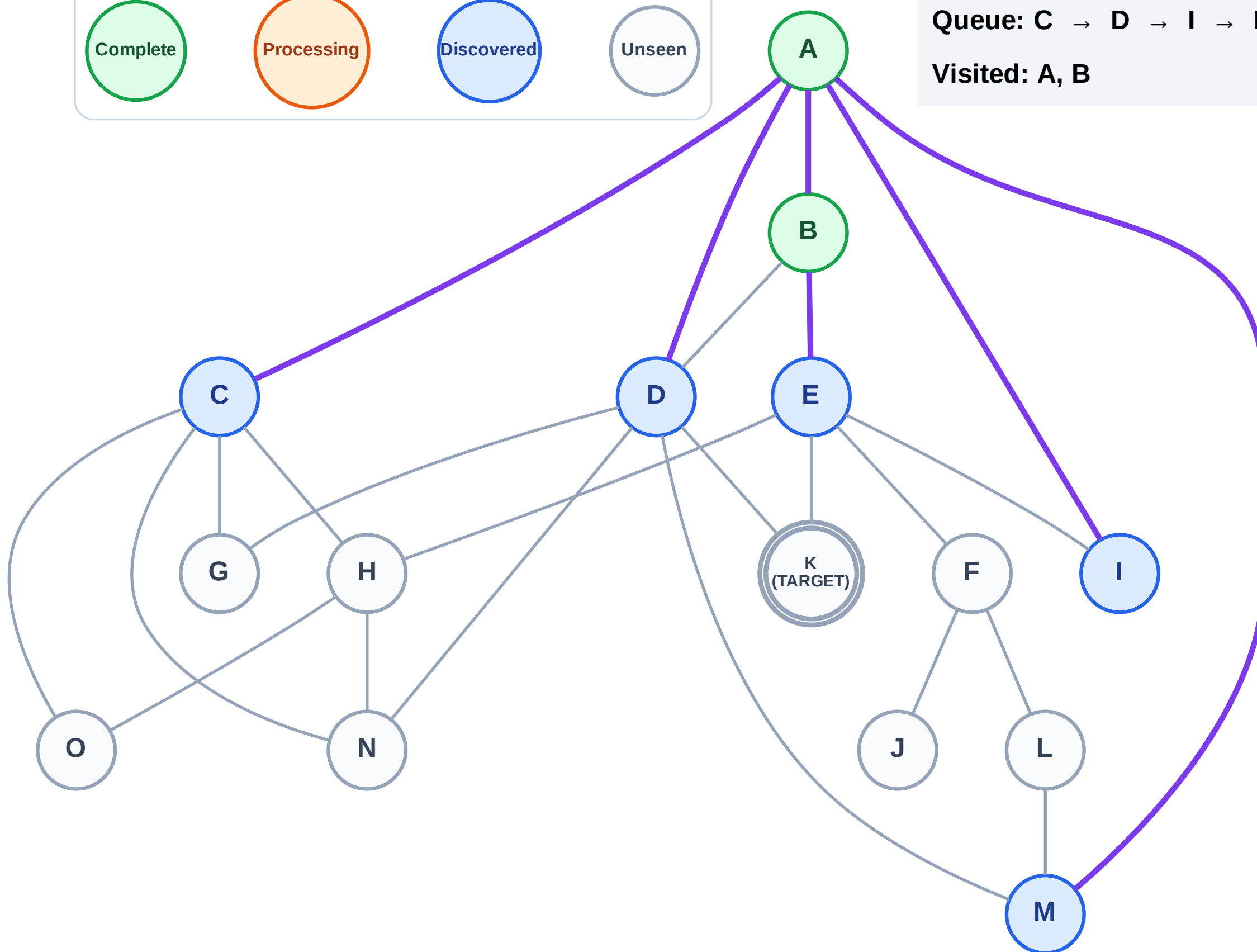
Step 5: Discover E from B

Legend



Queue: C → D → I → M → E

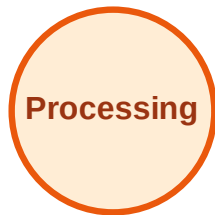
Visited: A, B



BFS Traversal

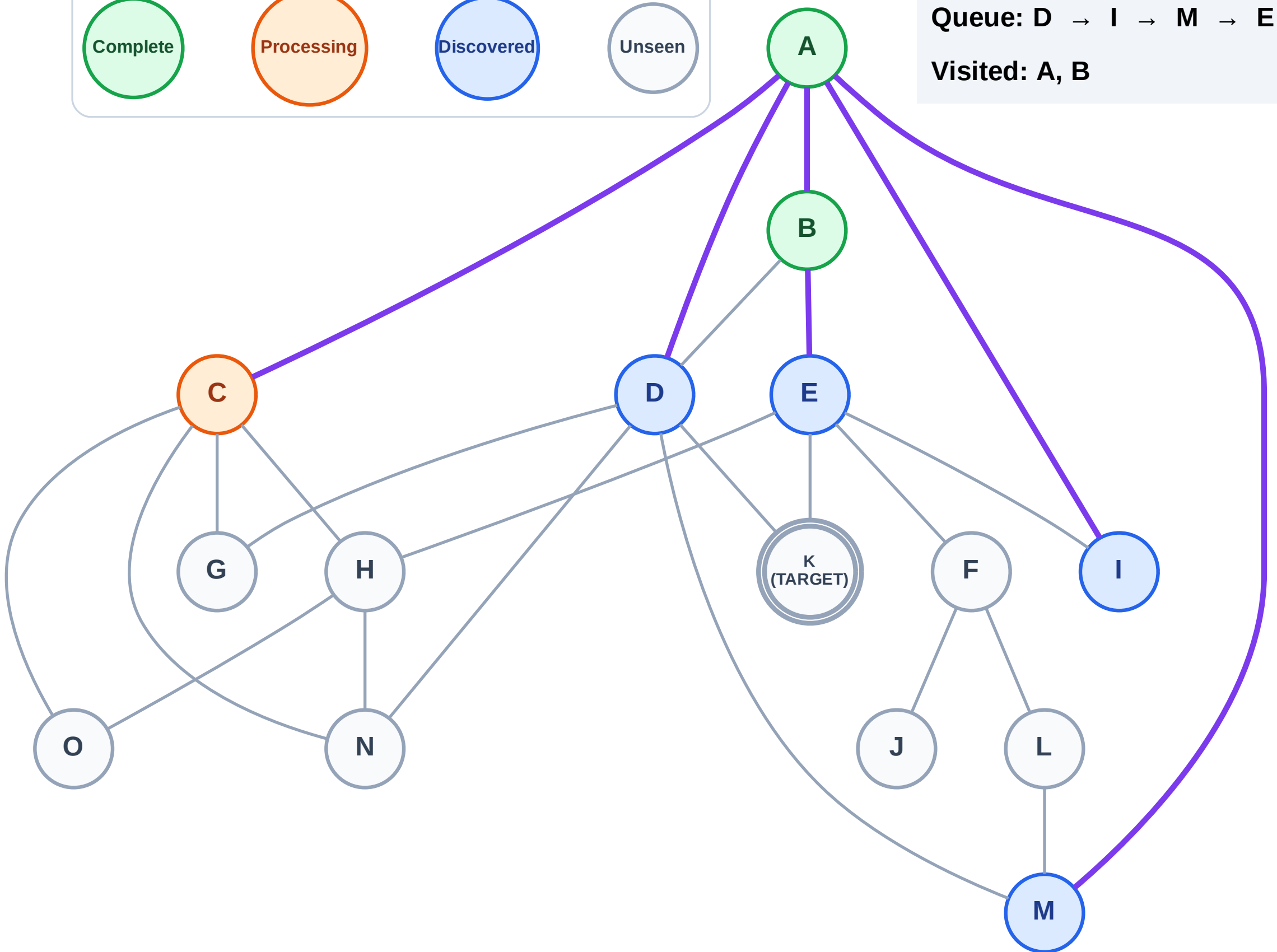
Step 6: Process node C

Legend



Queue: D → I → M → E

Visited: A, B



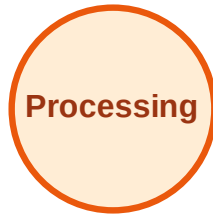
BFS Traversal

Step 7: Discover G, H, N, O from C

Legend



Complete



Processing



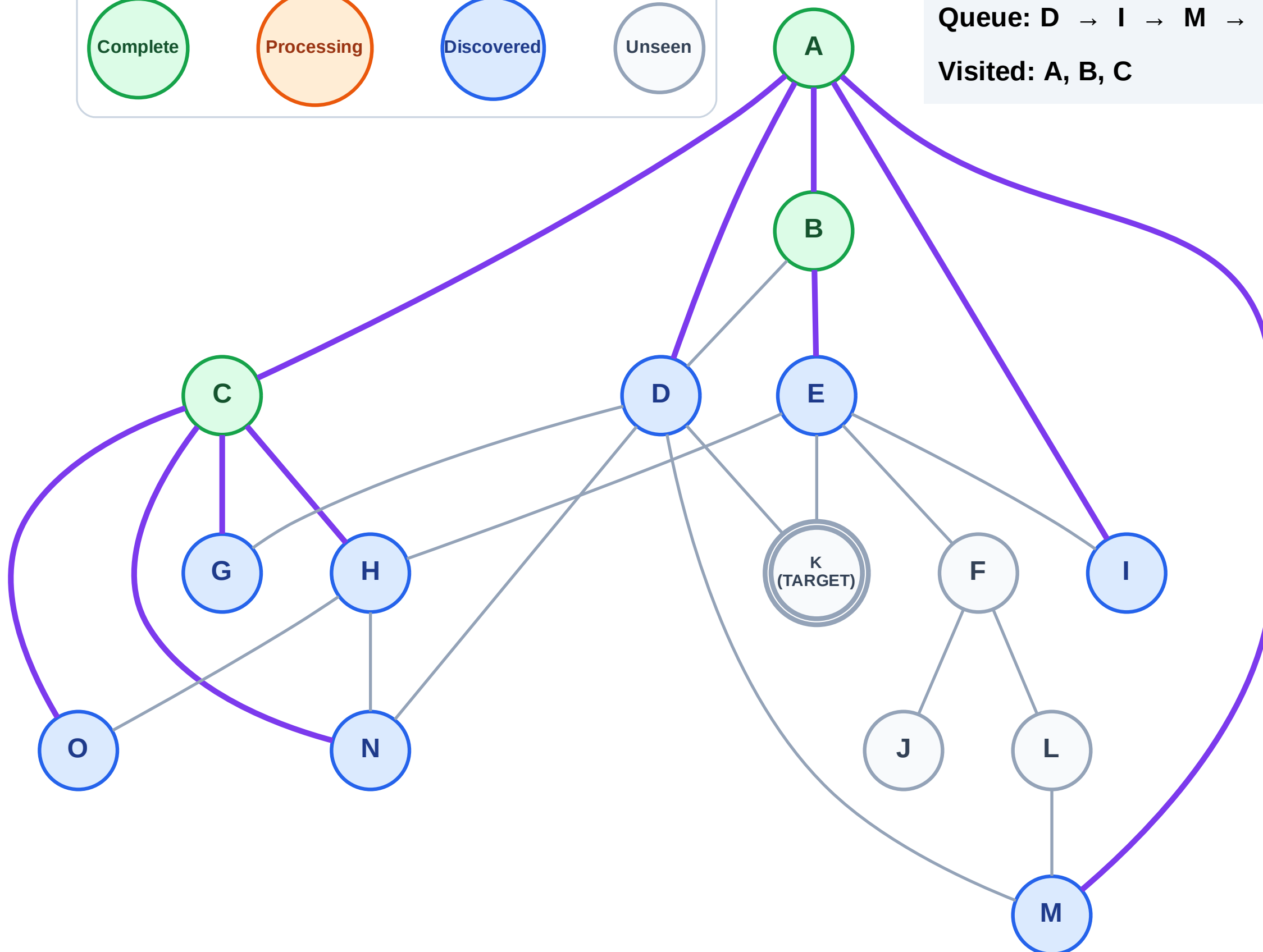
Discovered



Unseen

Queue: D → I → M → E → G → H → N → O

Visited: A, B, C



BFS Traversal

Step 8: Process node D

Legend

Complete

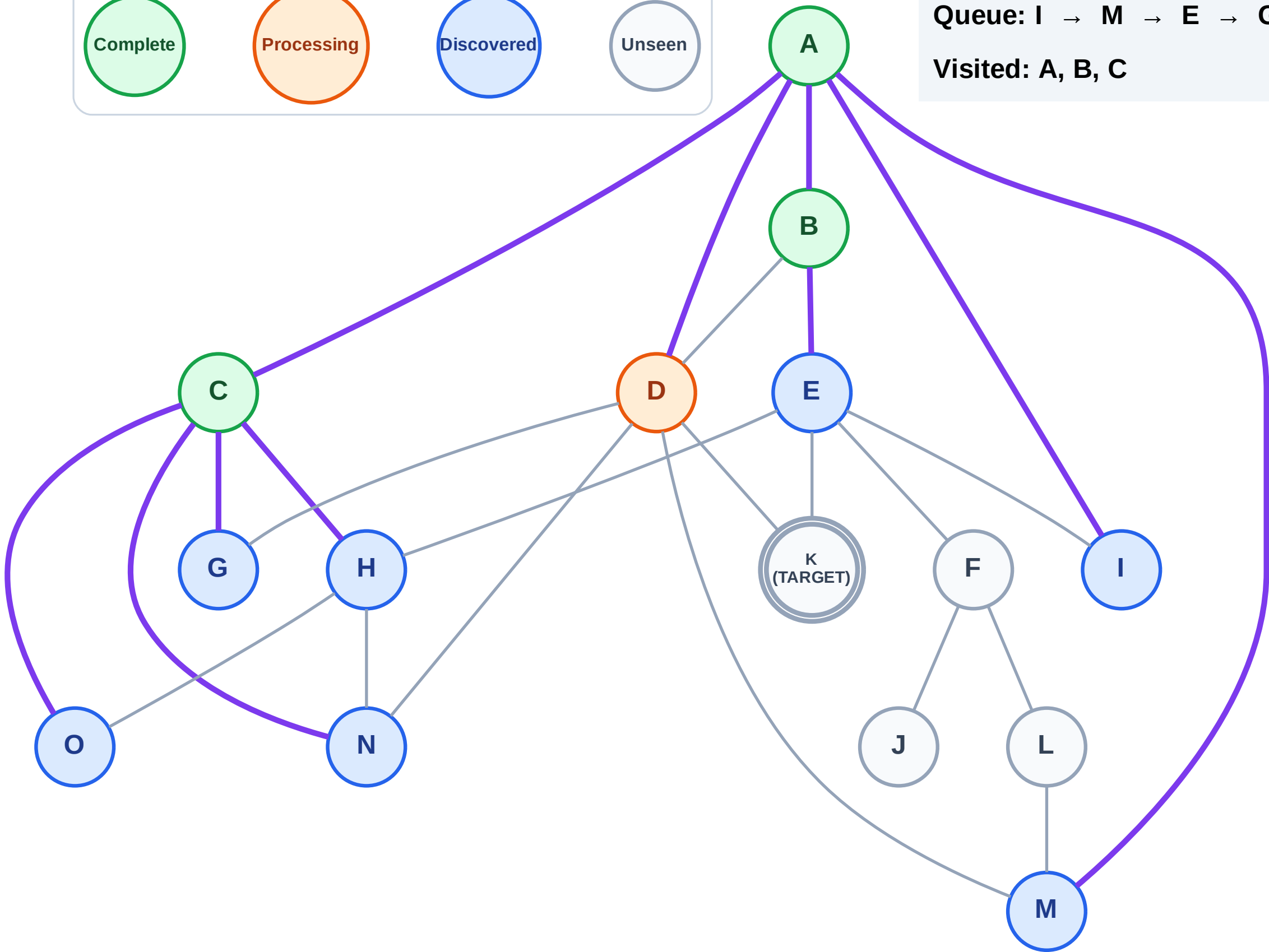
Processing

Discovered

Unseen

Queue: I → M → E → G → H → N → O

Visited: A, B, C



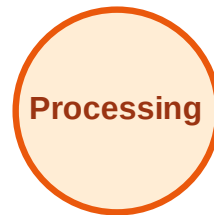
BFS Traversal

Step 9: Discover K from D

Legend



Complete



Processing



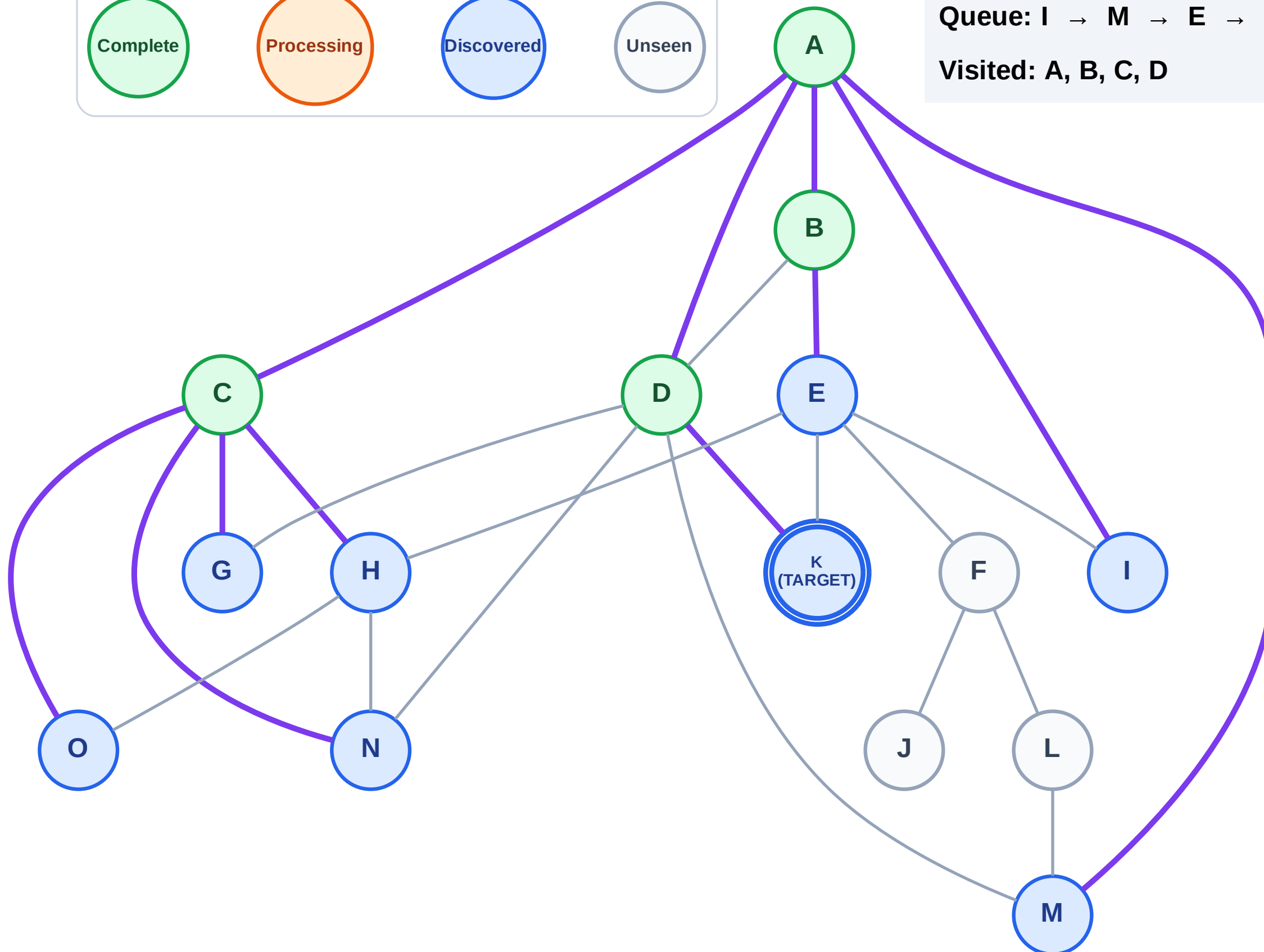
Discovered



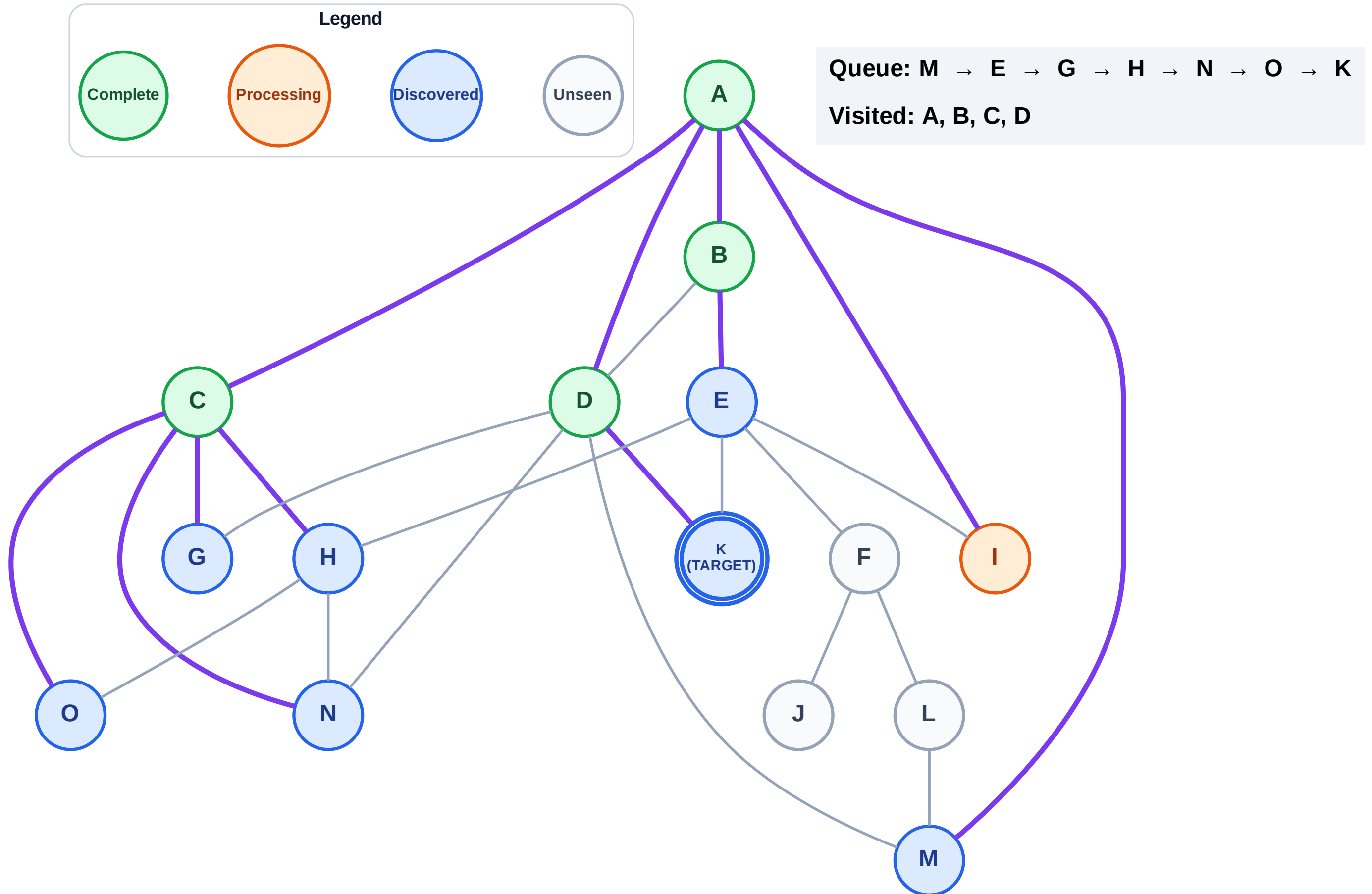
Unseen

Queue: I → M → E → G → H → N → O → K

Visited: A, B, C, D



Step 10: Process node I



BFS Traversal

Step 11: No new neighbors from I

Legend

Complete

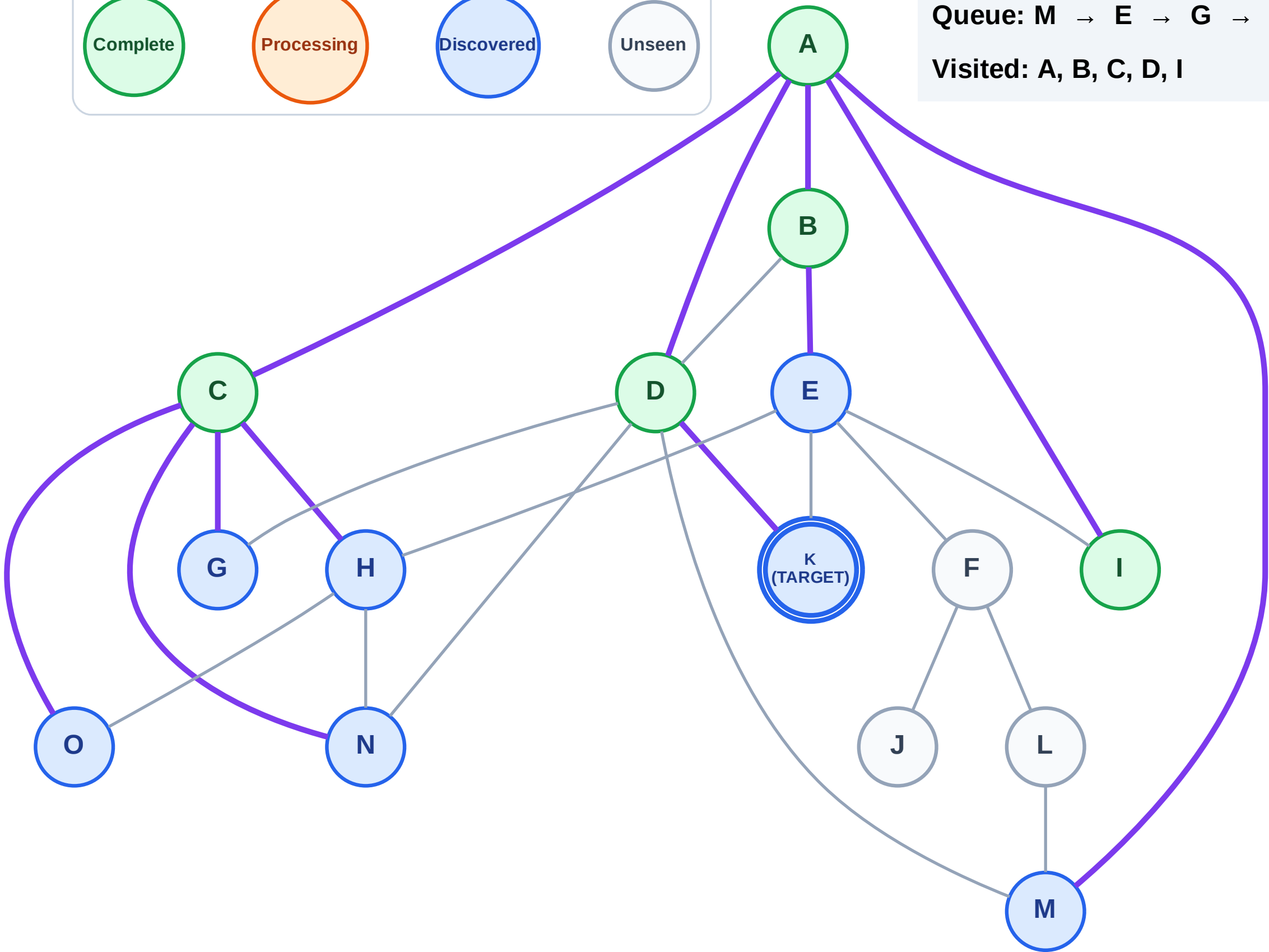
Processing

Discovered

Unseen

Queue: M → E → G → H → N → O → K

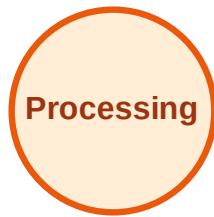
Visited: A, B, C, D, I



BFS Traversal

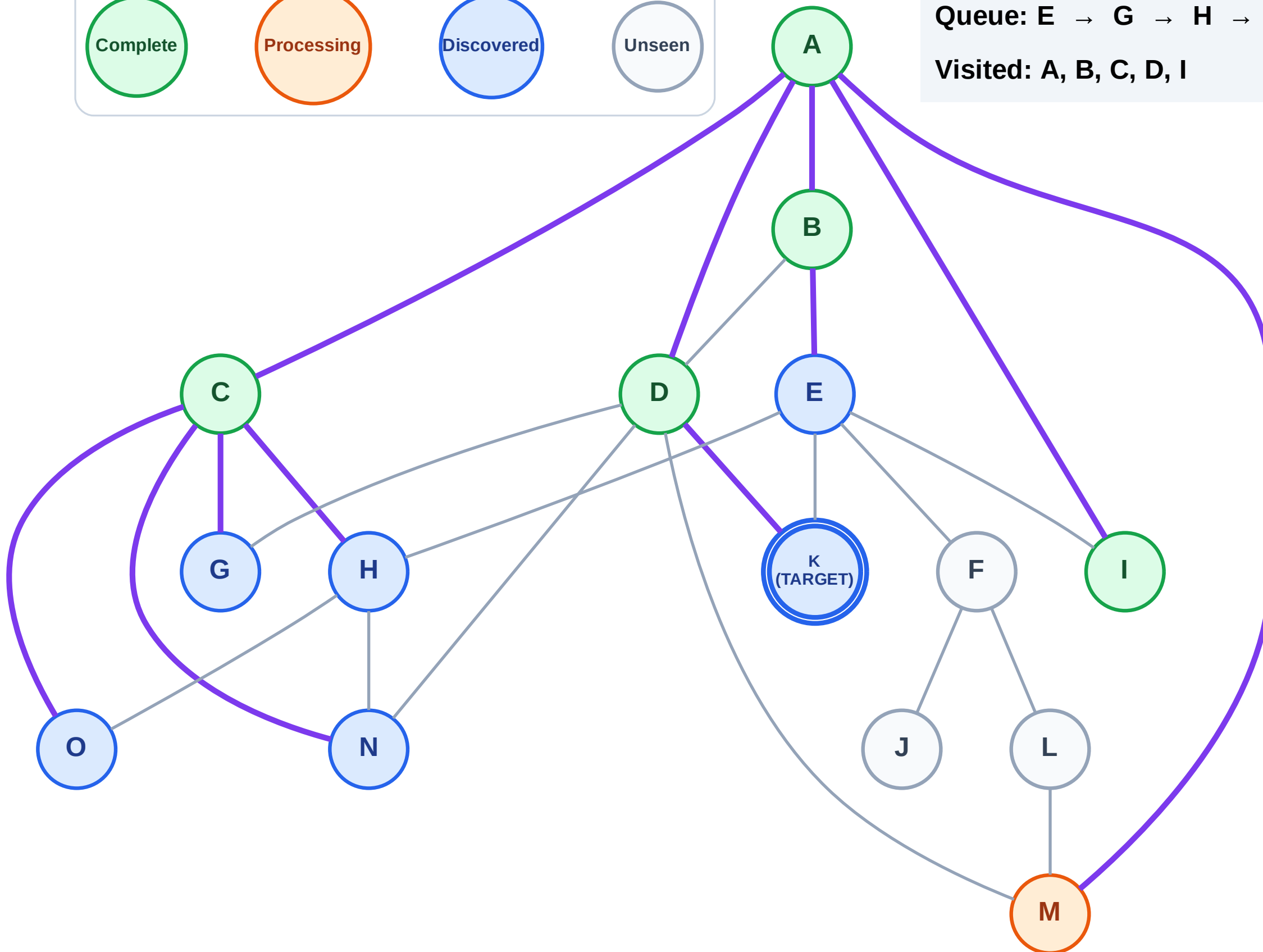
Step 12: Process node M

Legend



Queue: E → G → H → N → O → K

Visited: A, B, C, D, I



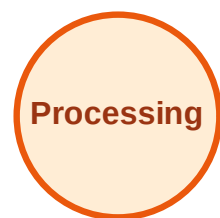
BFS Traversal

Step 13: Discover L from M

Legend



Complete



Processing



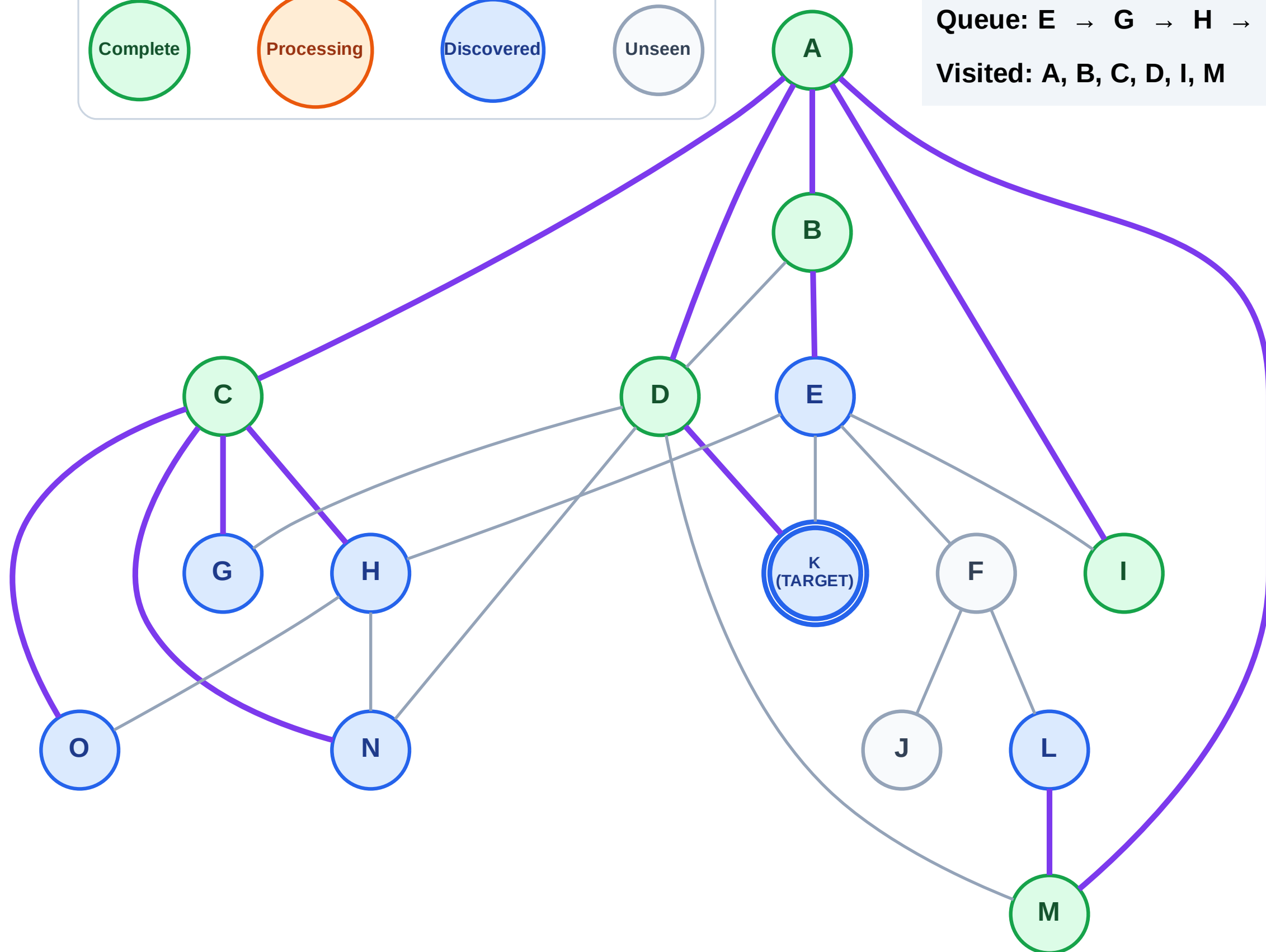
Discovered



Unseen

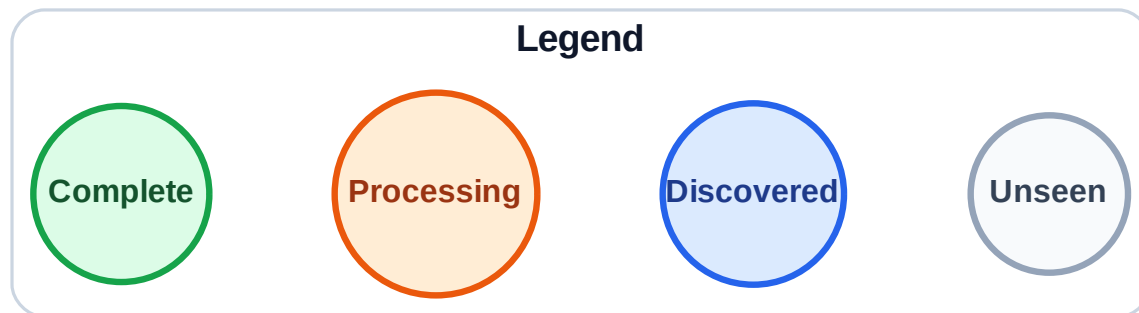
Queue: E → G → H → N → O → K → L

Visited: A, B, C, D, I, M



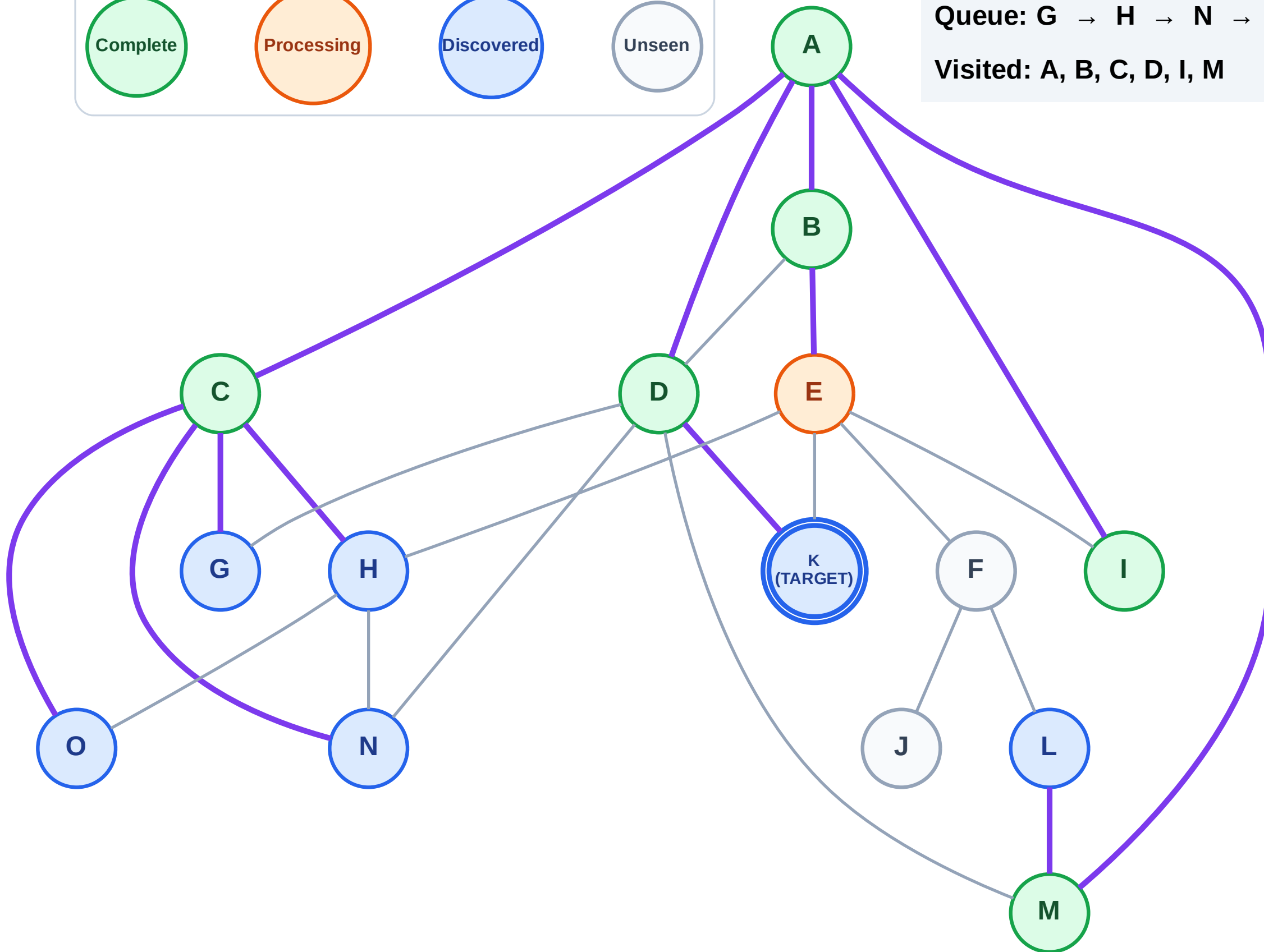
Step 14: Process node E

Step 14: Process node E



Queue: G → H → N → O → K → L

Visited: A, B, C, D, I, M



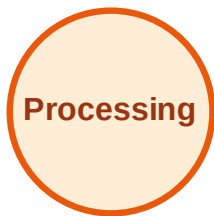
BFS Traversal

Step 15: Discover F from E

Legend



Complete



Processing



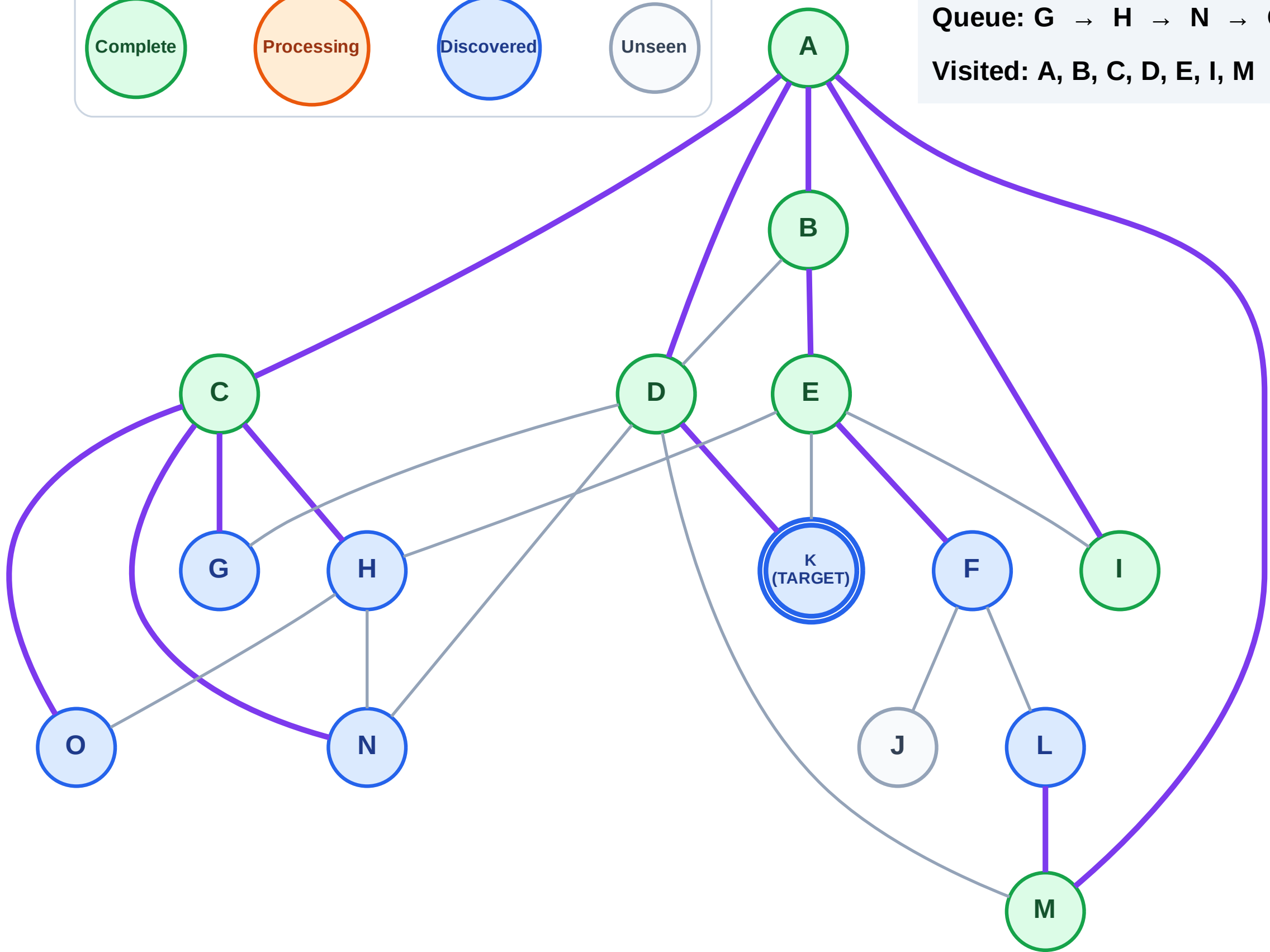
Discovered



Unseen

Queue: G → H → N → O → K → L → F

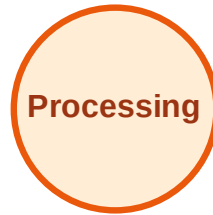
Visited: A, B, C, D, E, I, M



BFS Traversal

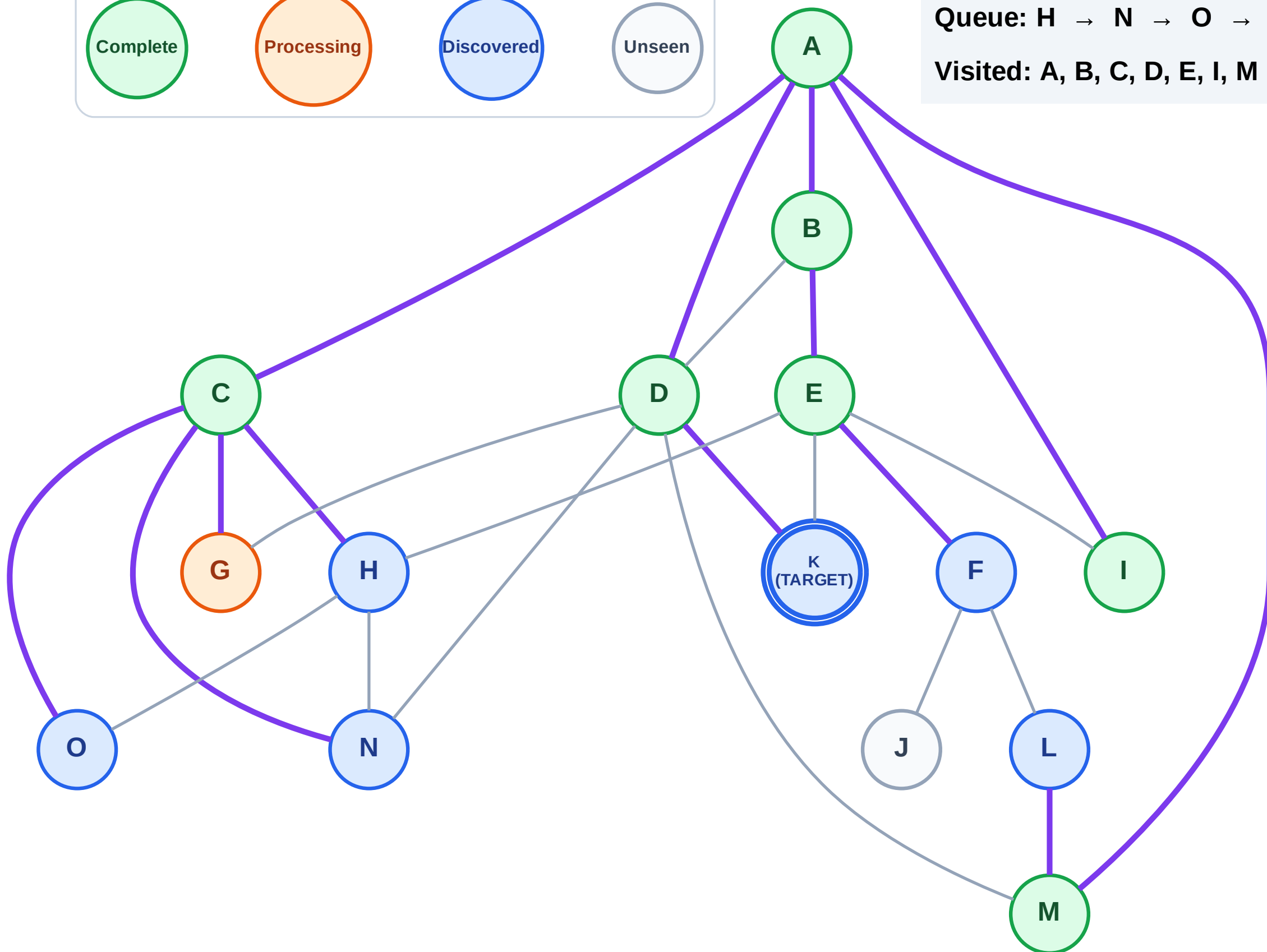
Step 16: Process node G

Legend



Queue: H → N → O → K → L → F

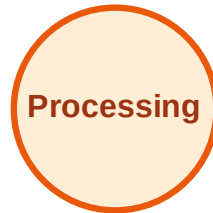
Visited: A, B, C, D, E, I, M



BFS Traversal

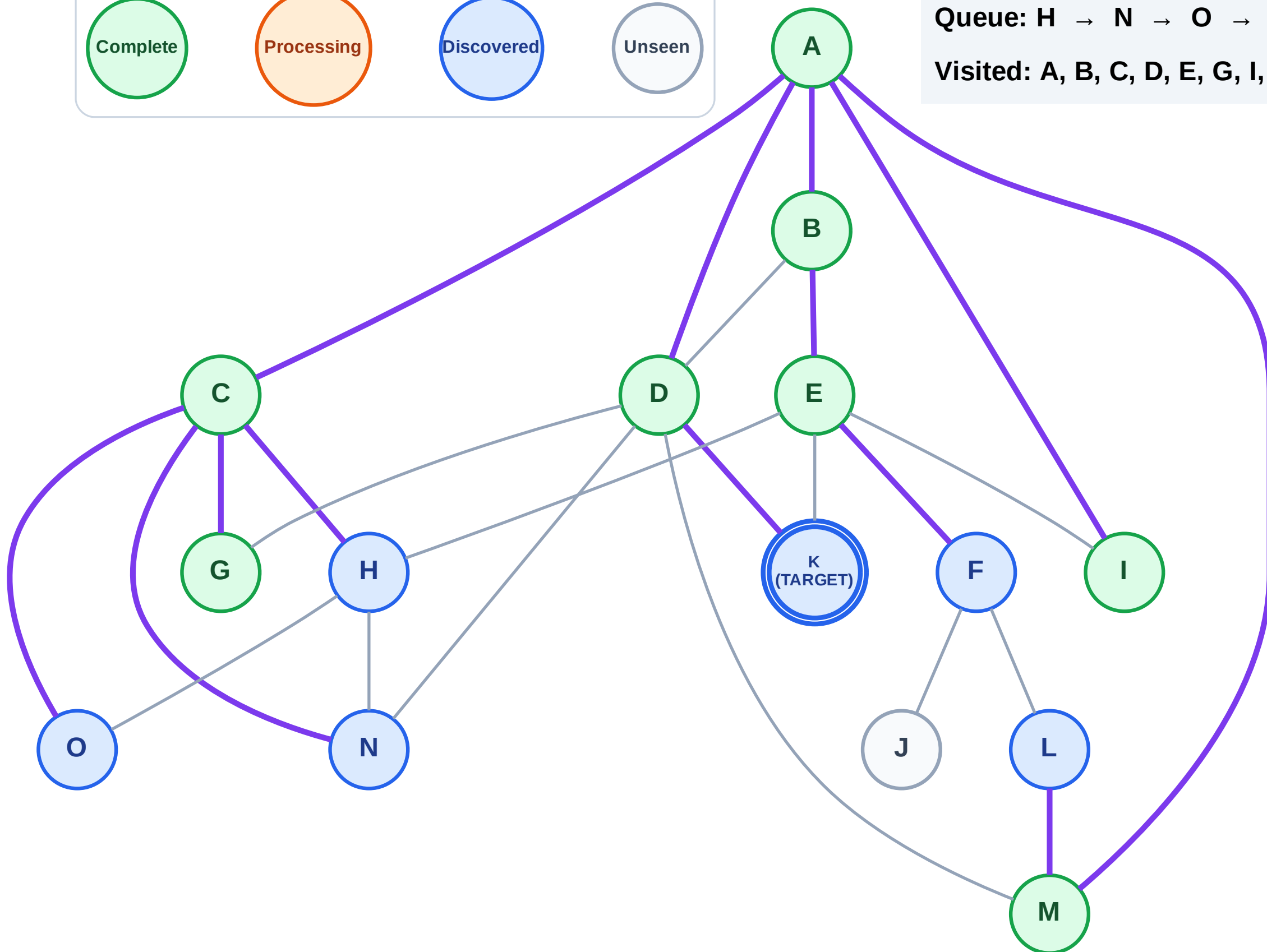
Step 17: No new neighbors from G

Legend



Queue: H → N → O → K → L → F

Visited: A, B, C, D, E, G, I, M



BFS Traversal

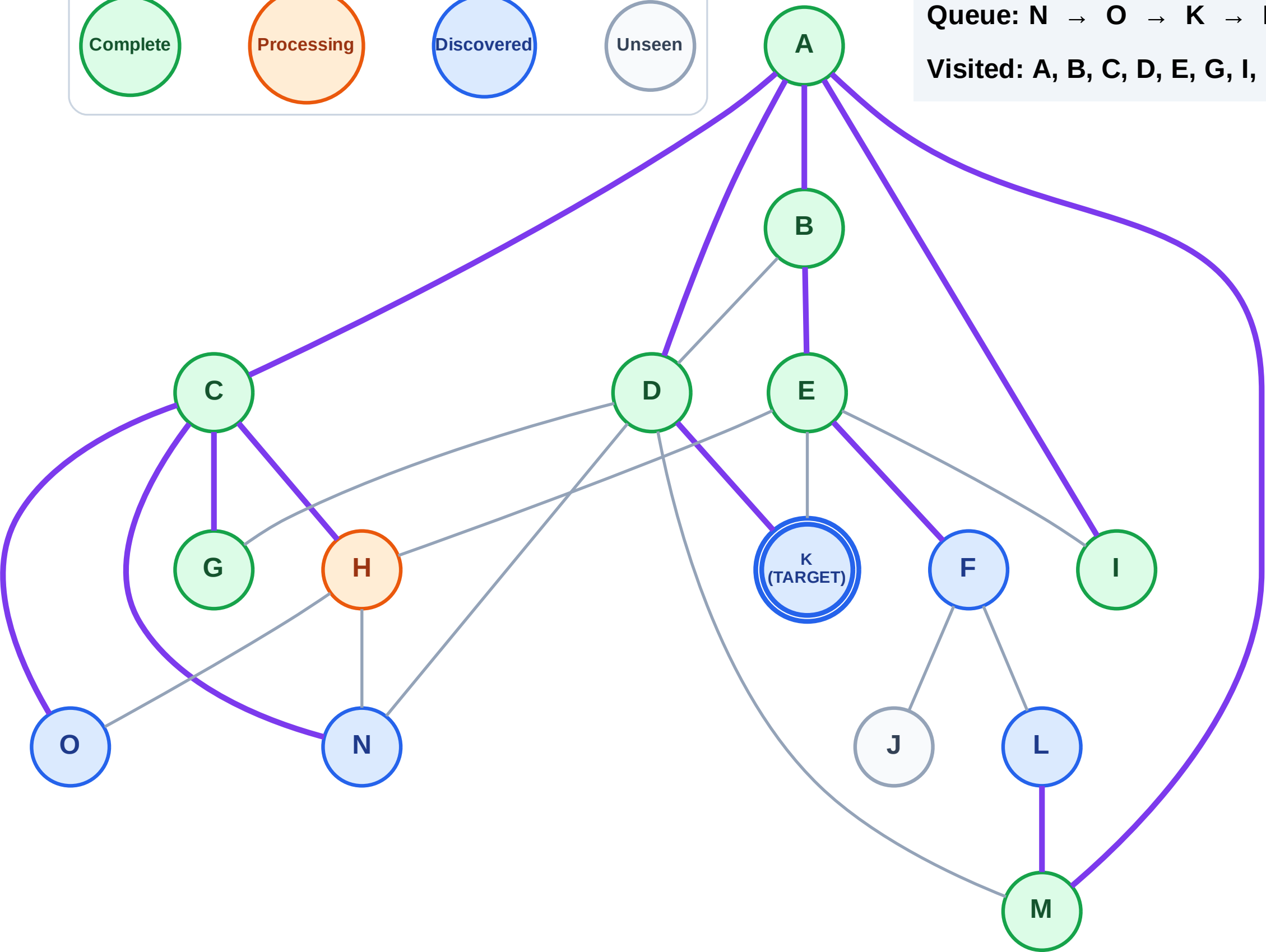
Step 18: Process node H

Legend



Queue: N → O → K → L → F

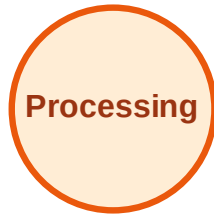
Visited: A, B, C, D, E, G, I, M



BFS Traversal

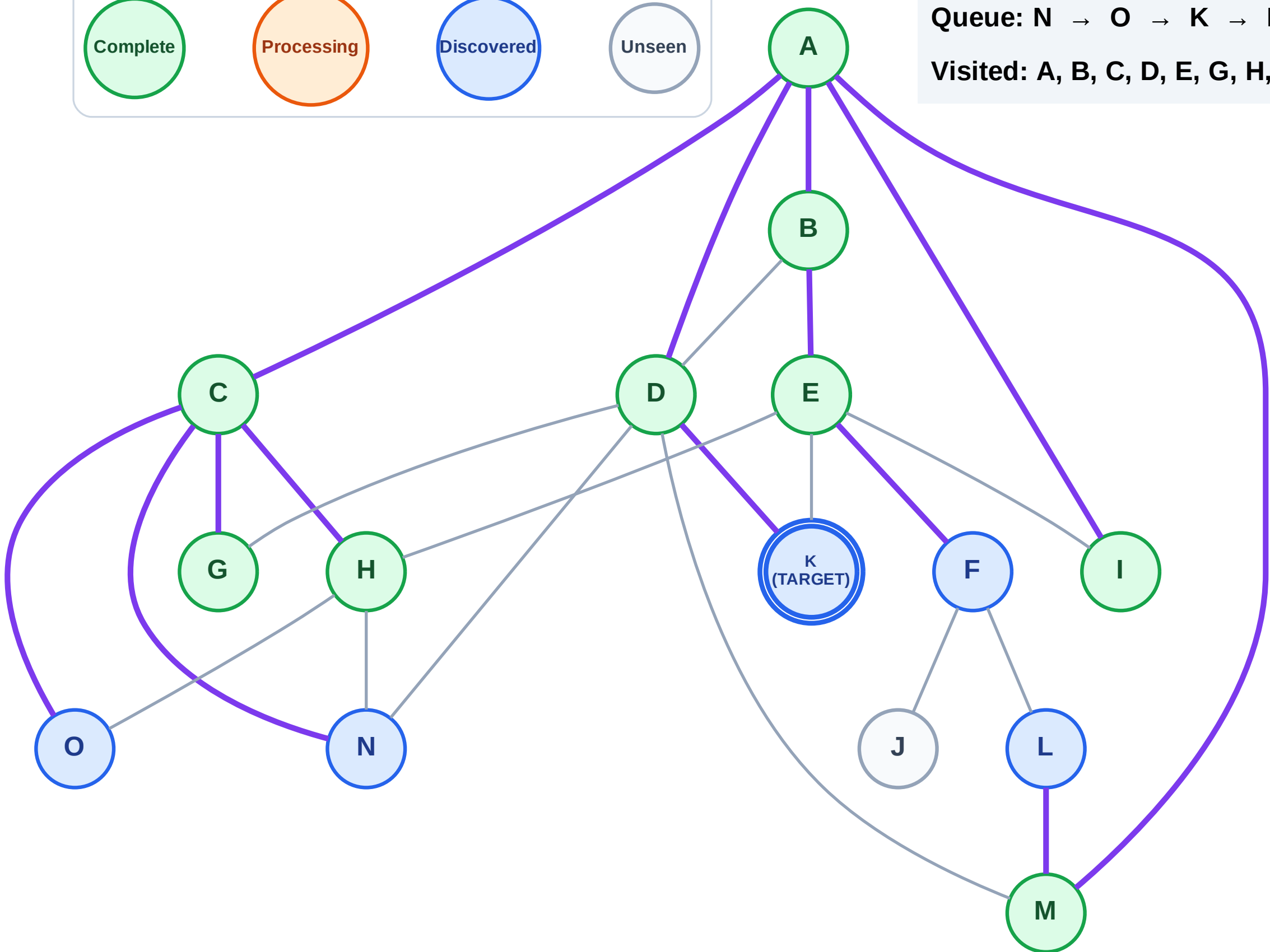
Step 19: No new neighbors from H

Legend



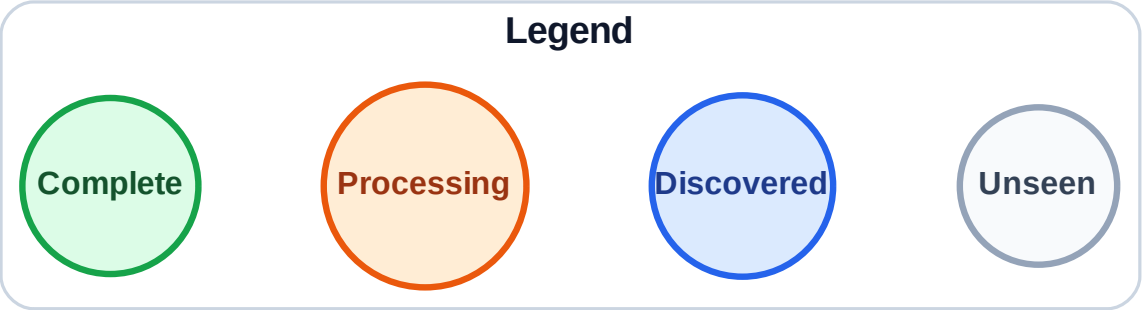
Queue: N → O → K → L → F

Visited: A, B, C, D, E, G, H, I, M



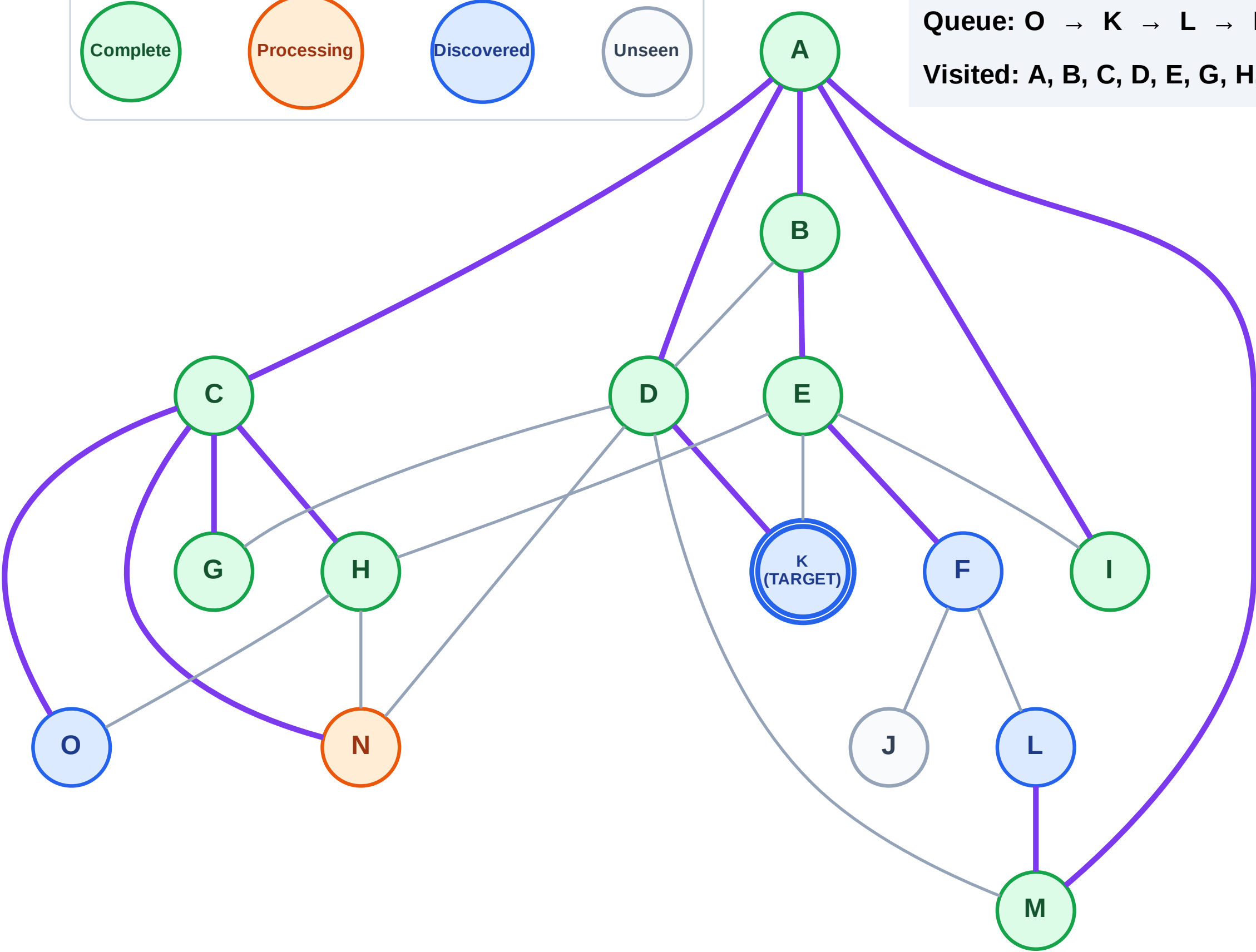
BFS Traversal

Step 20: Process node N



Queue: O → K → L → F

Visited: A, B, C, D, E, G, H, I, M

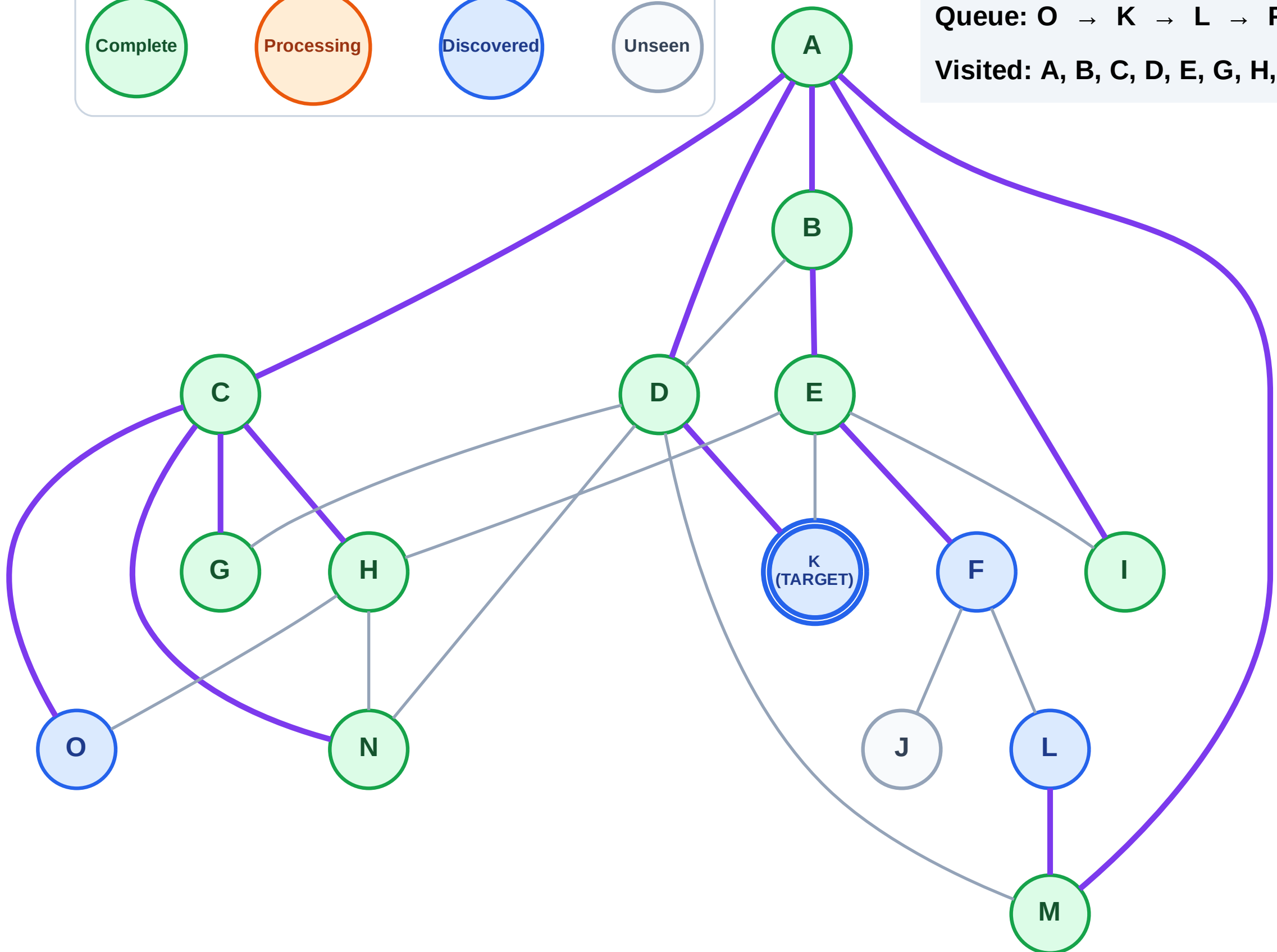


Step 21: No new neighbors from N

Legend



Visited: A, B, C, D, E, G, H, I, M, N

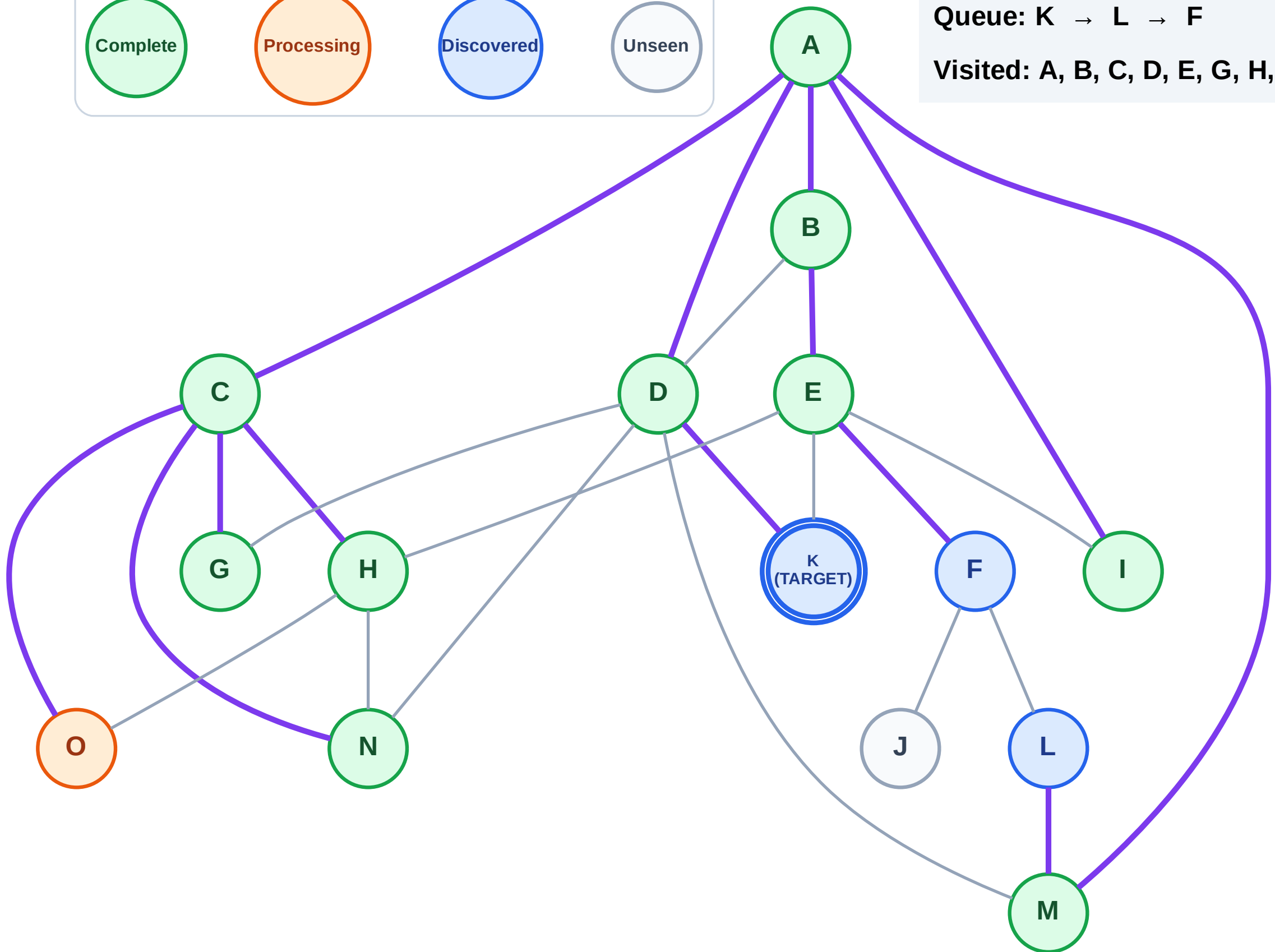


Step 22: Process node O

Legend



Visited: A, B, C, D, E, G, H, I, M, N

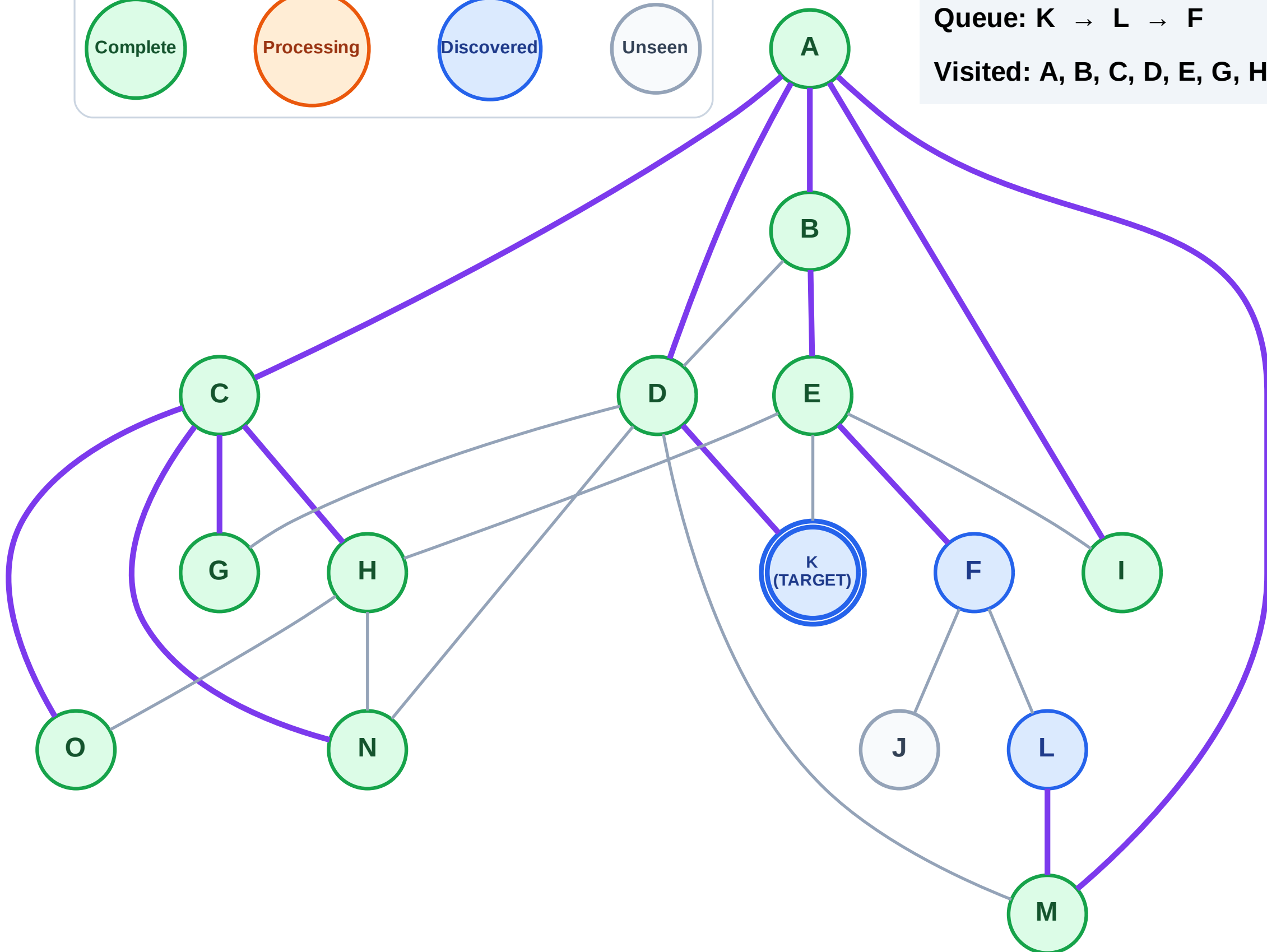


Step 23: No new neighbors from O

Legend

Unseen

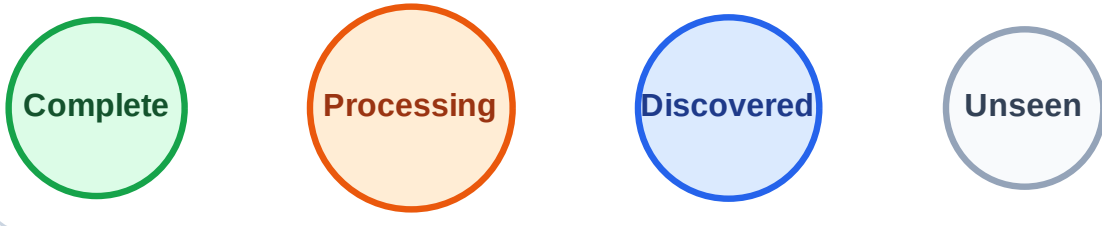
Visited: A, B, C, D, E, G, H, I, M, N, O



BFS Traversal

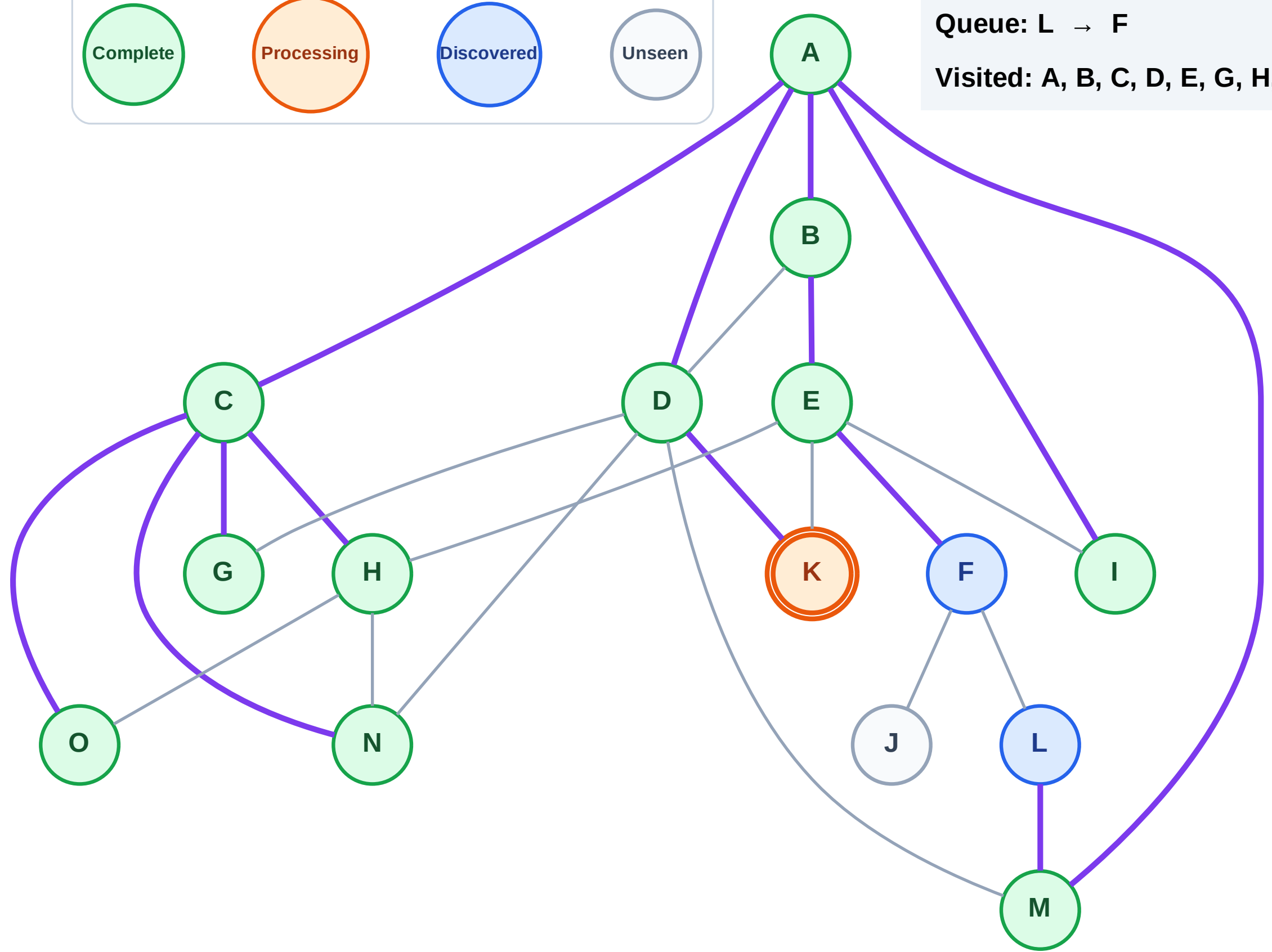
Step 24: Process node K

Legend



Queue: L → F

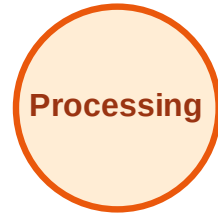
Visited: A, B, C, D, E, G, H, I, M, N, O



BFS Traversal

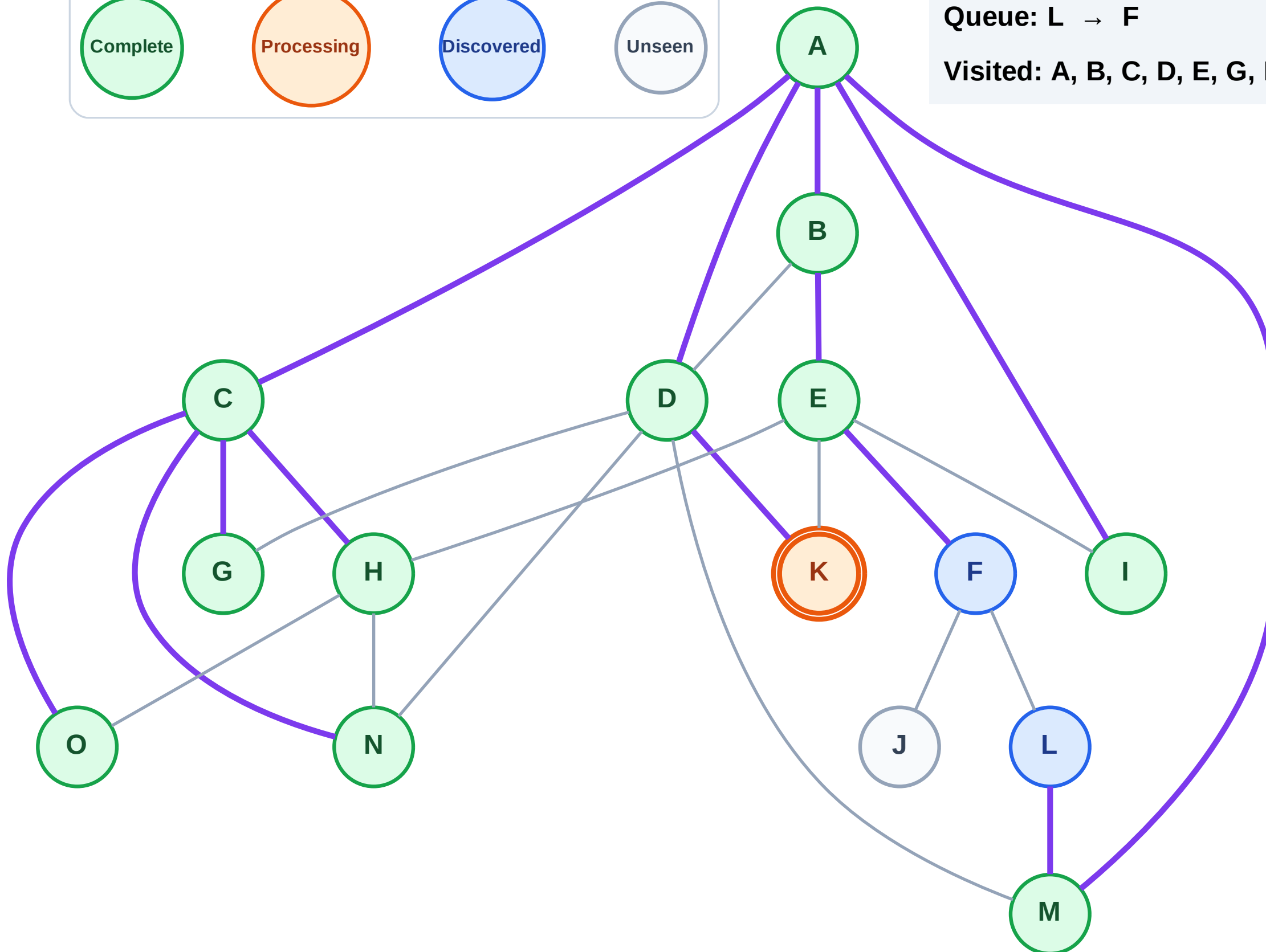
Step 25: TARGET NODE K REACHED! Path (2 hops): A → D → K

Legend



Queue: L → F

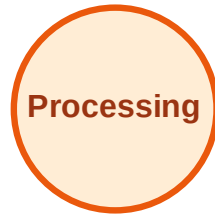
Visited: A, B, C, D, E, G, H, I, K, M, N, O



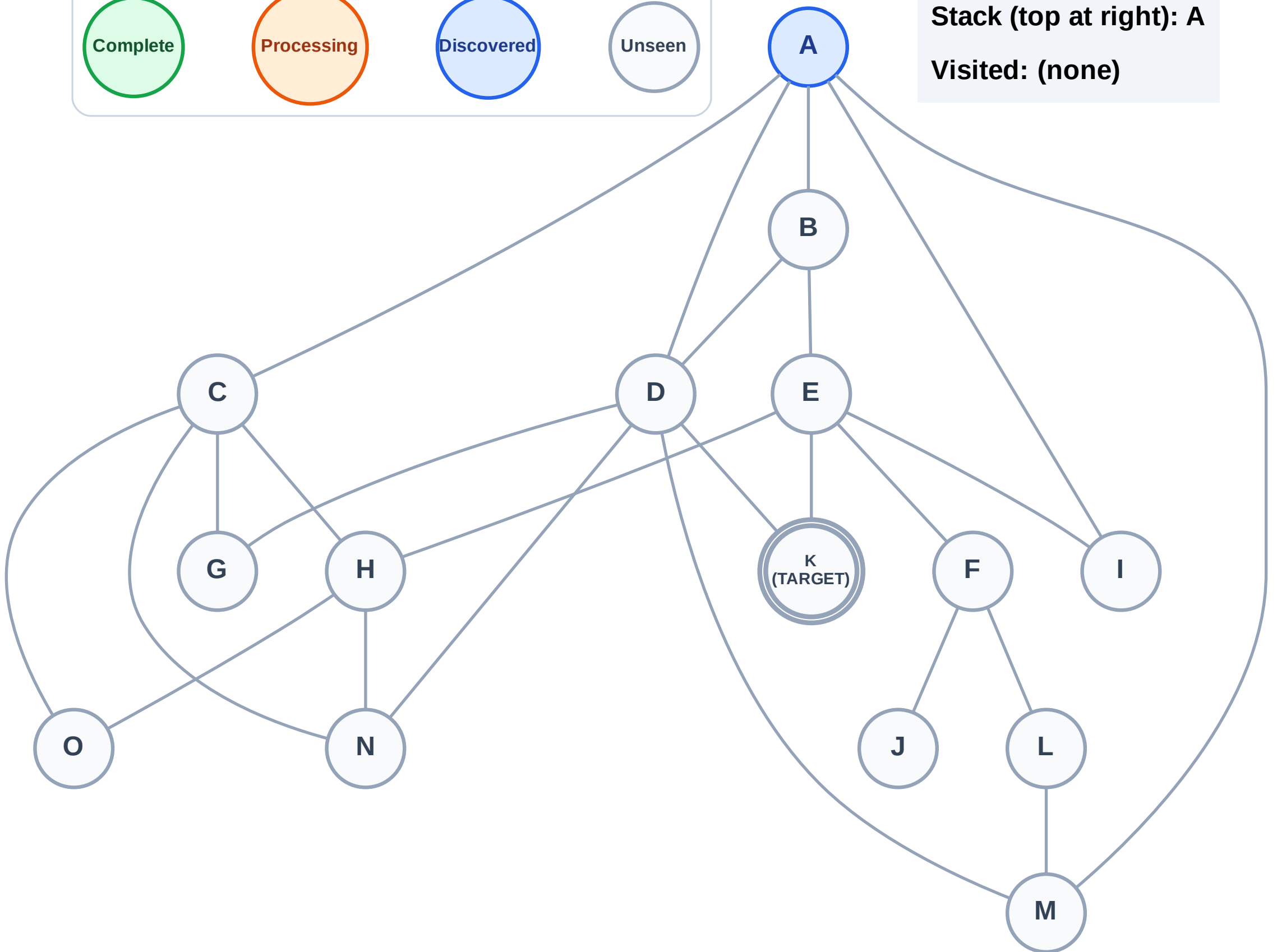
DFS Traversal

Step 1: Start at A (Seeking target node K)

Legend



Stack (top at right): A
Visited: (none)



DFS Traversal

Step 2: Process node A

Legend

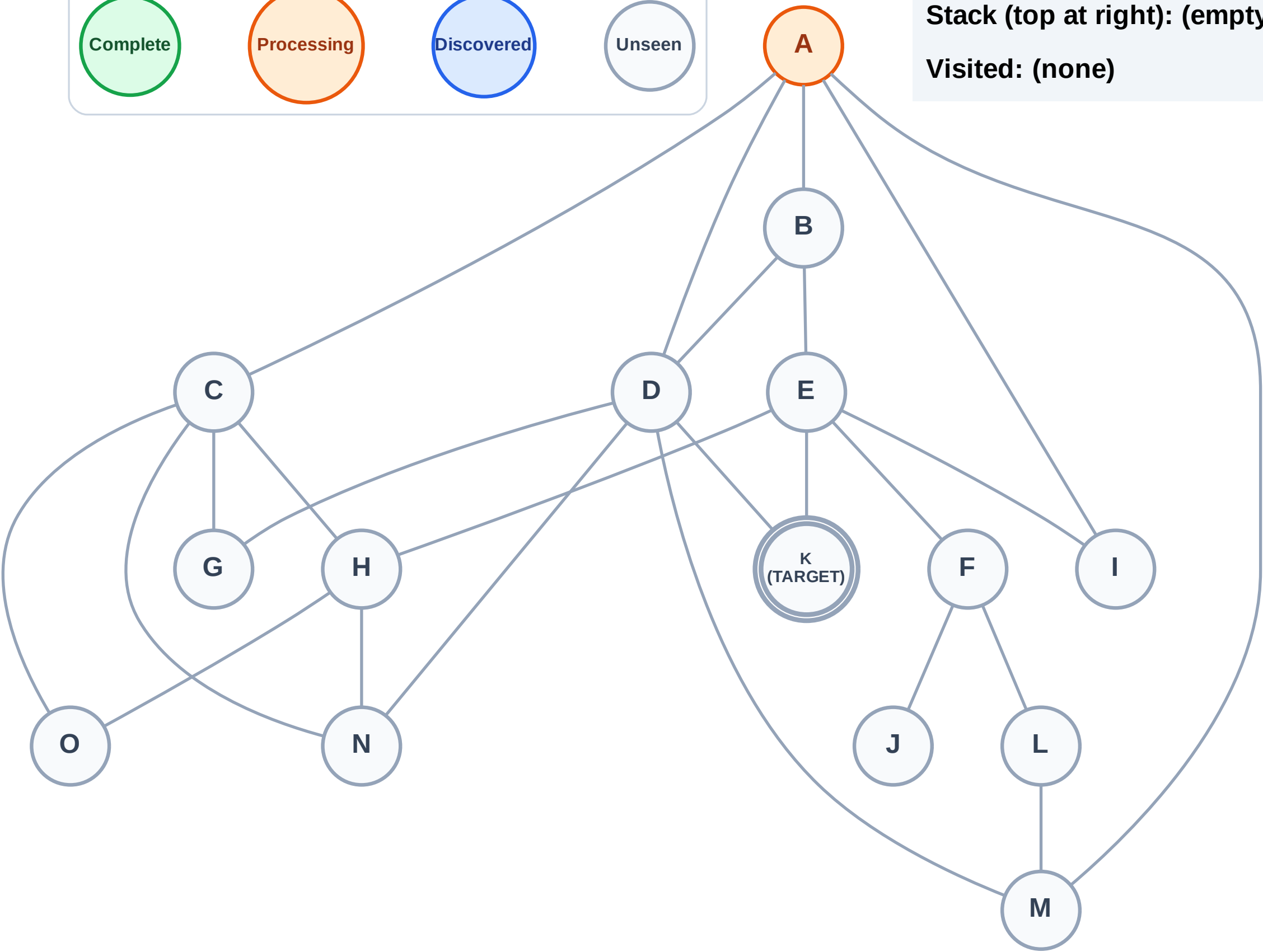
Complete

Processing

Discovered

Unseen

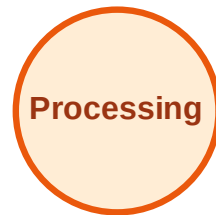
Stack (top at right): (empty)
Visited: (none)



DFS Traversal

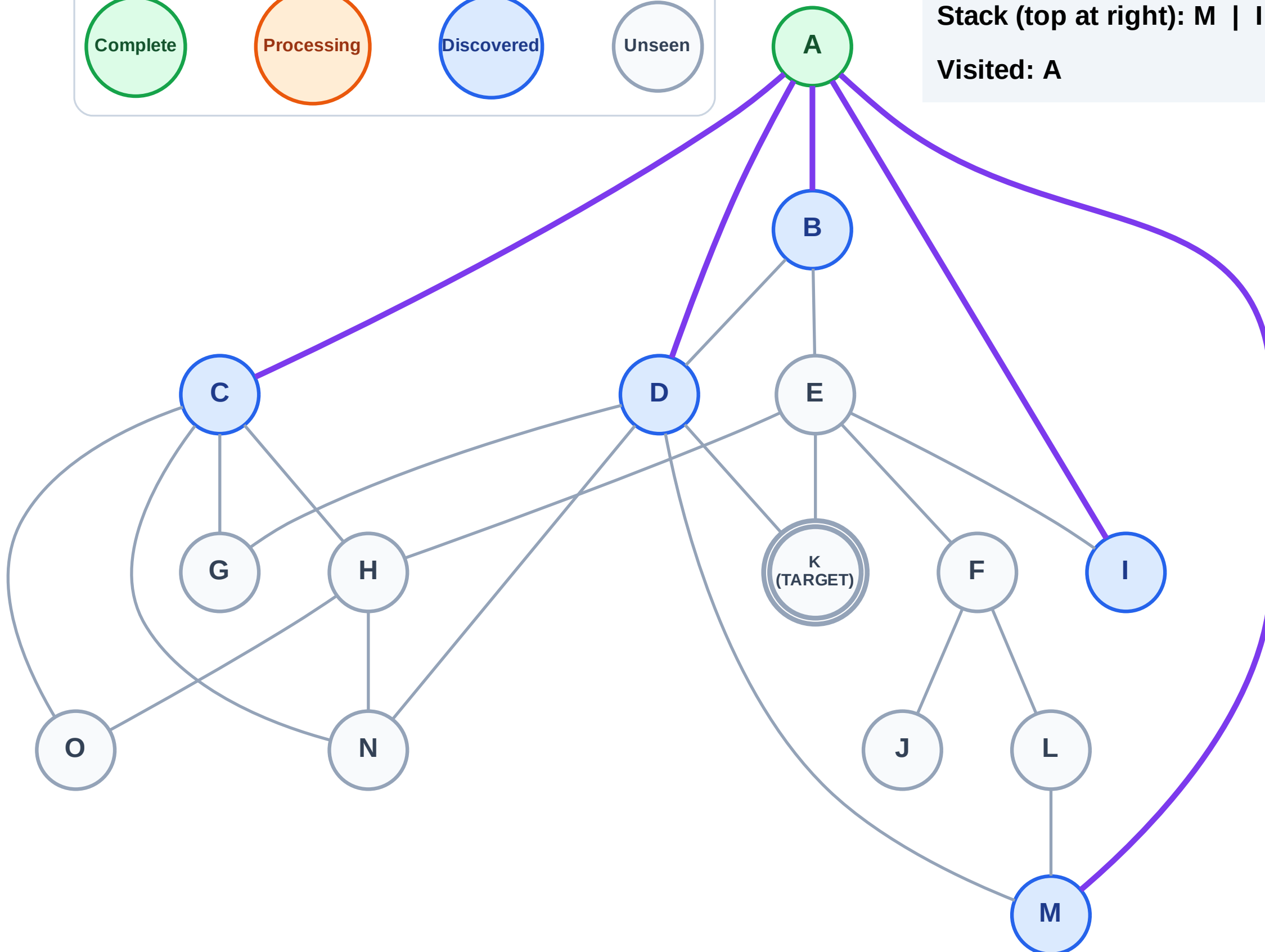
Step 3: Discover M, I, D, C, B from A

Legend



Stack (top at right): M | I | D | C | B

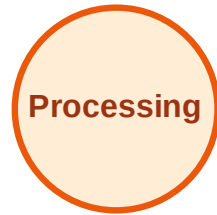
Visited: A



DFS Traversal

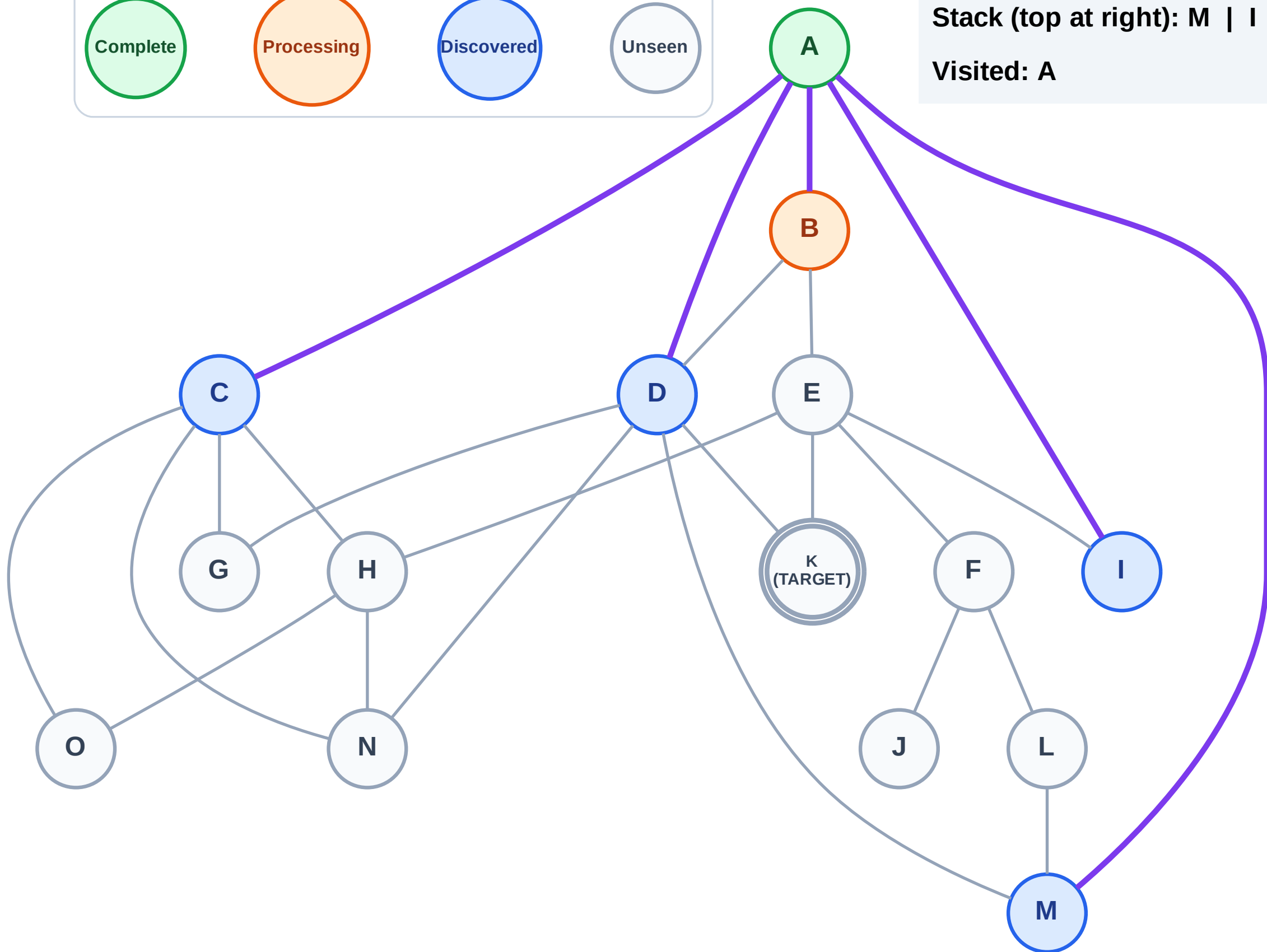
Step 4: Process node B

Legend



Stack (top at right): M | I | D | C

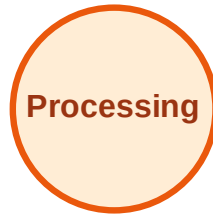
Visited: A



DFS Traversal

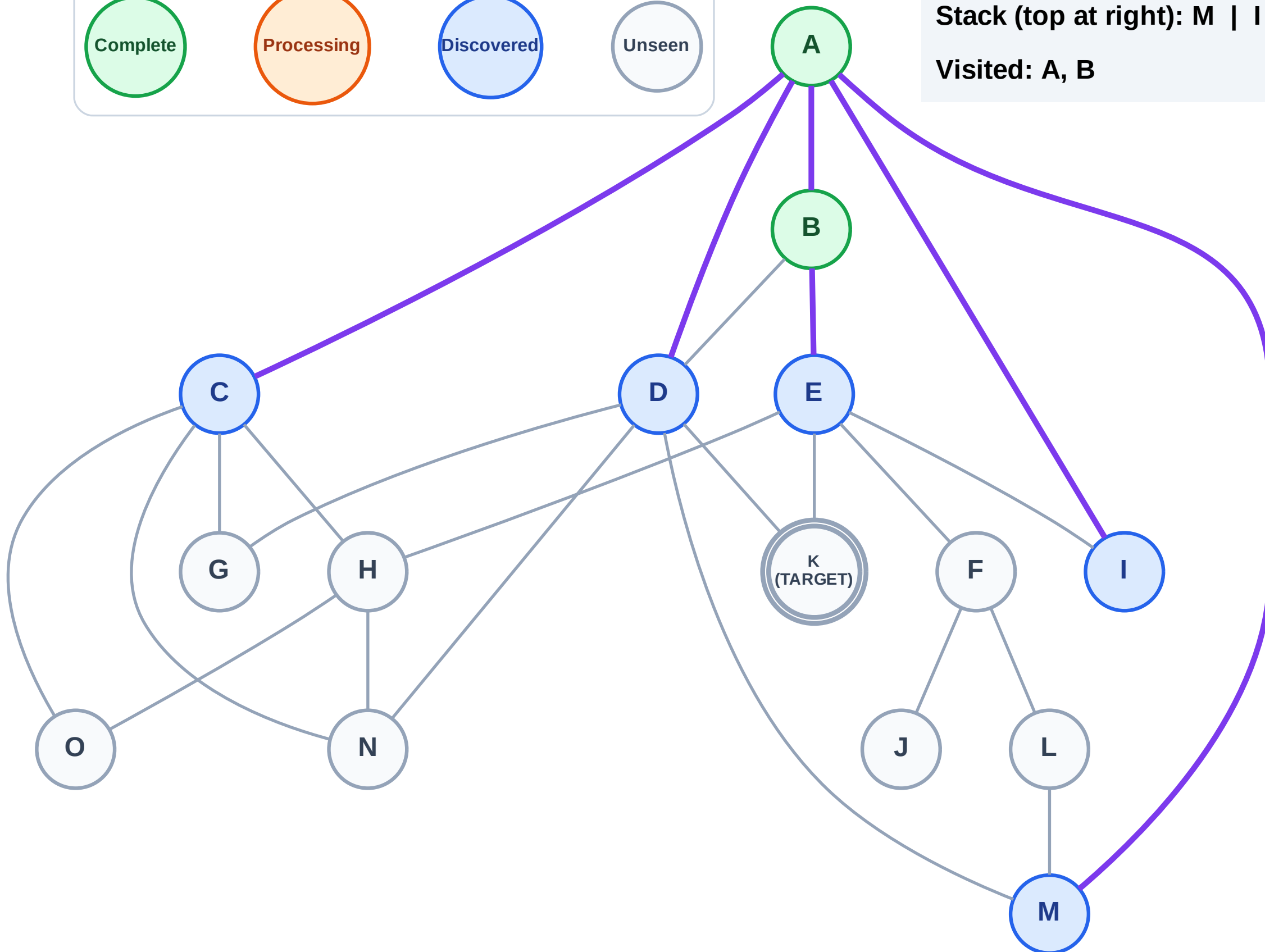
Step 5: Discover E from B

Legend



Stack (top at right): M | I | D | C | E

Visited: A, B



DFS Traversal

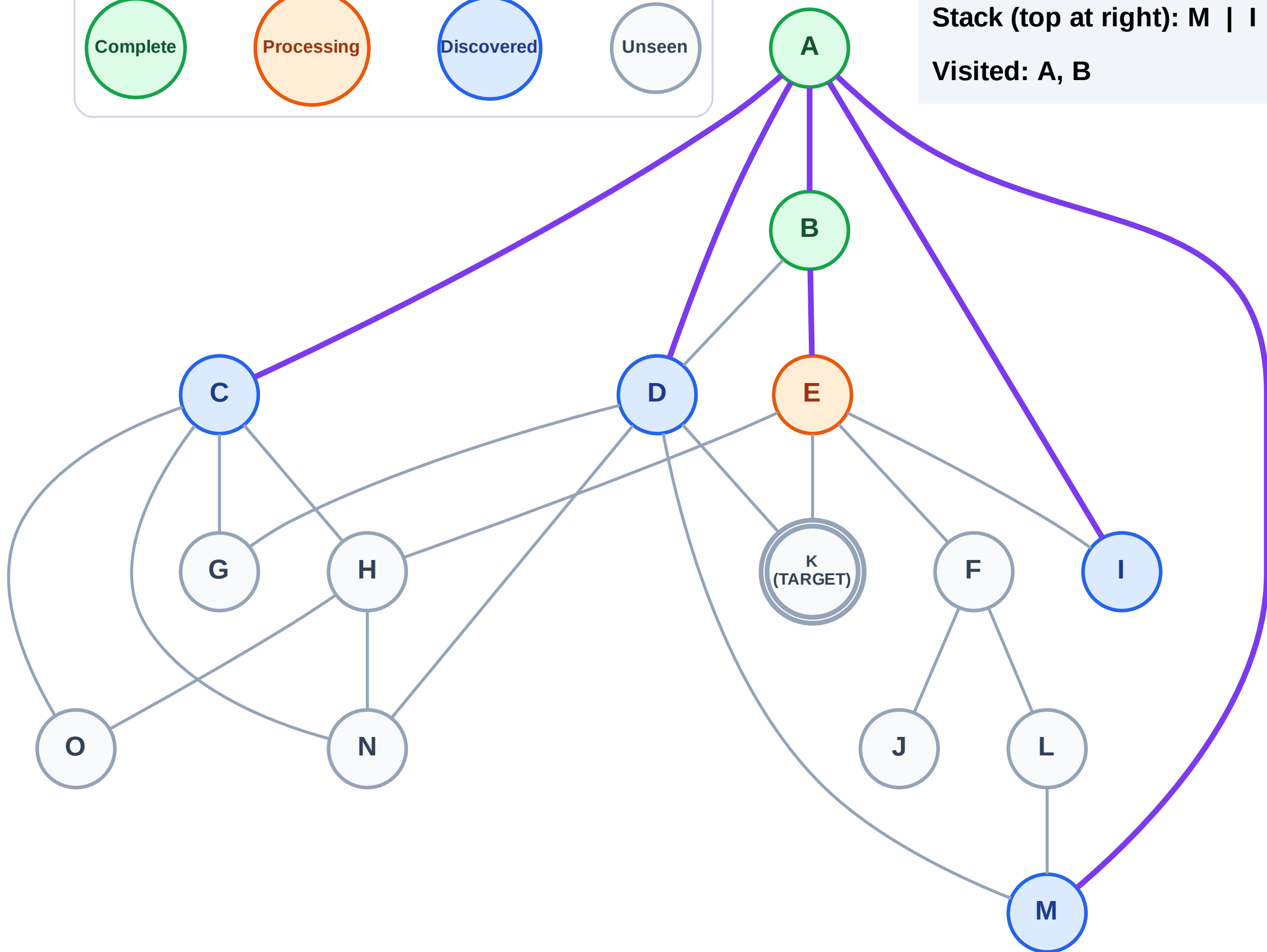
Step 6: Process node E

Legend



Stack (top at right): M | I | D | C

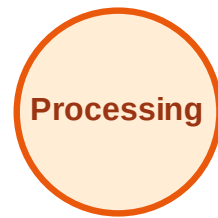
Visited: A, B



DFS Traversal

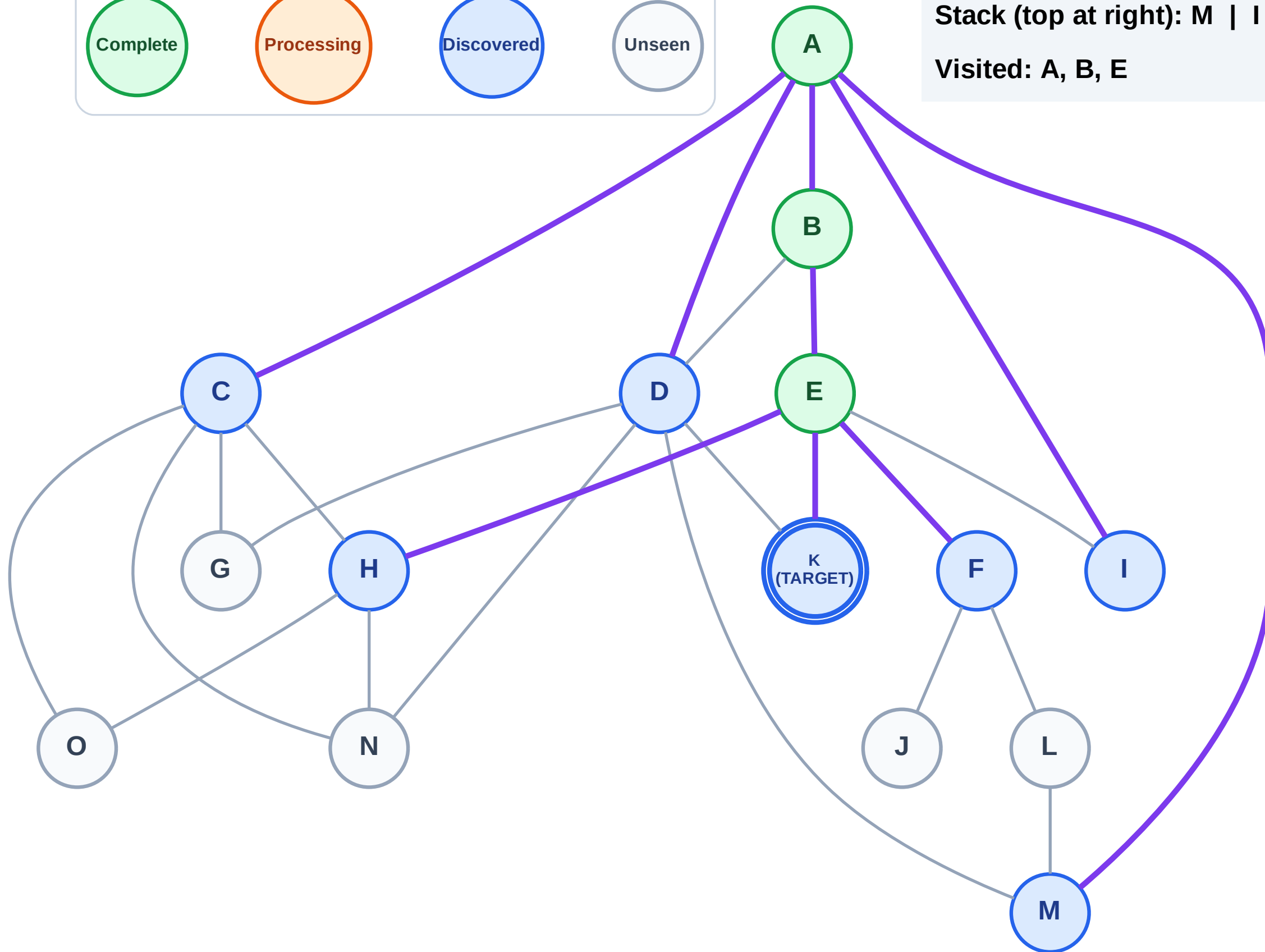
Step 7: Discover K, H, F from E

Legend



Stack (top at right): M | I | D | C | K | H | F

Visited: A, B, E



DFS Traversal

Step 8: Process node F

Legend

Complete

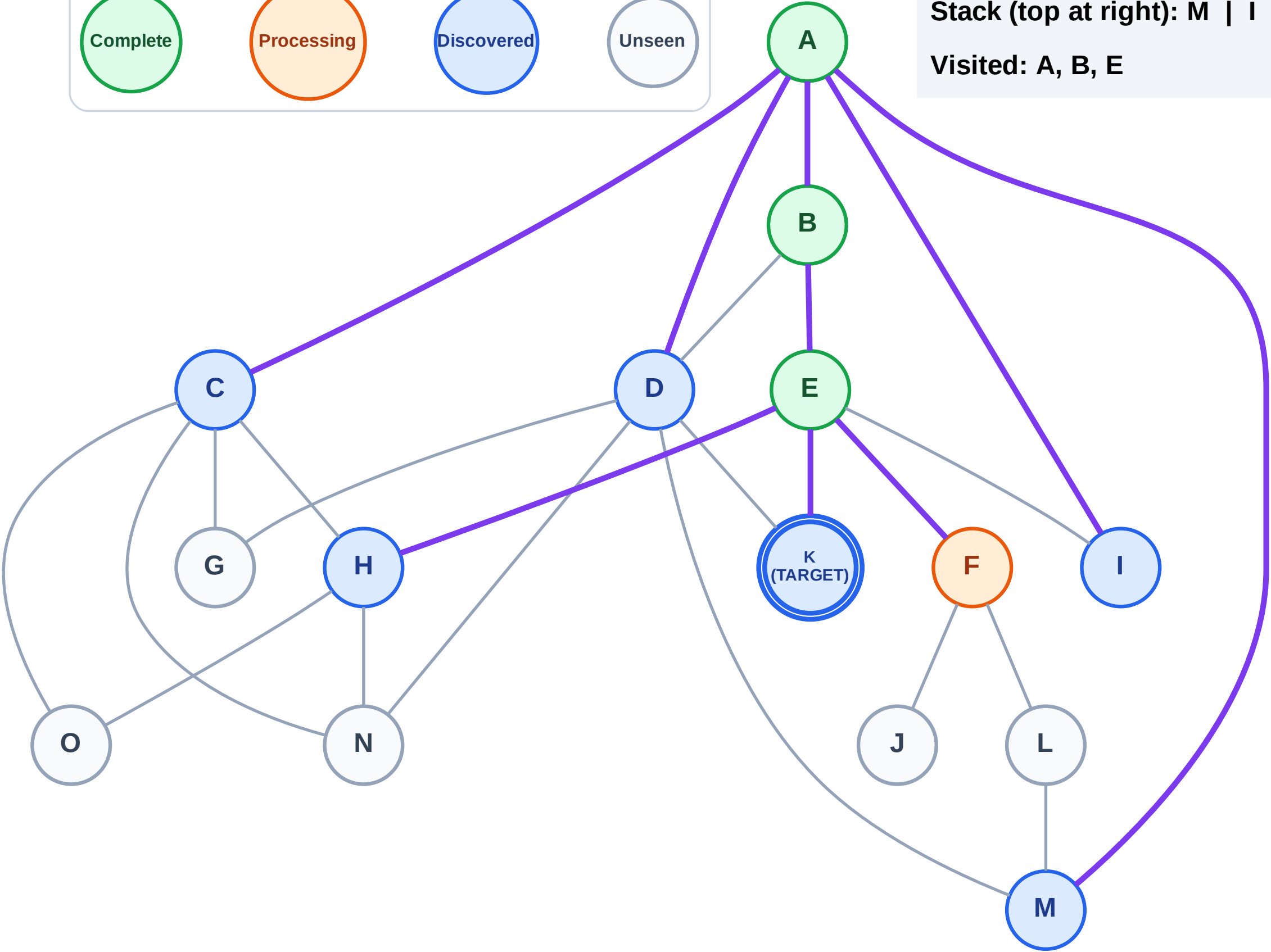
Processing

Discovered

Unseen

Stack (top at right): M | I | D | C | K | H

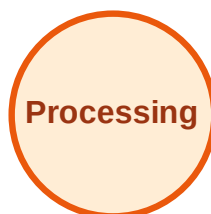
Visited: A, B, E



DFS Traversal

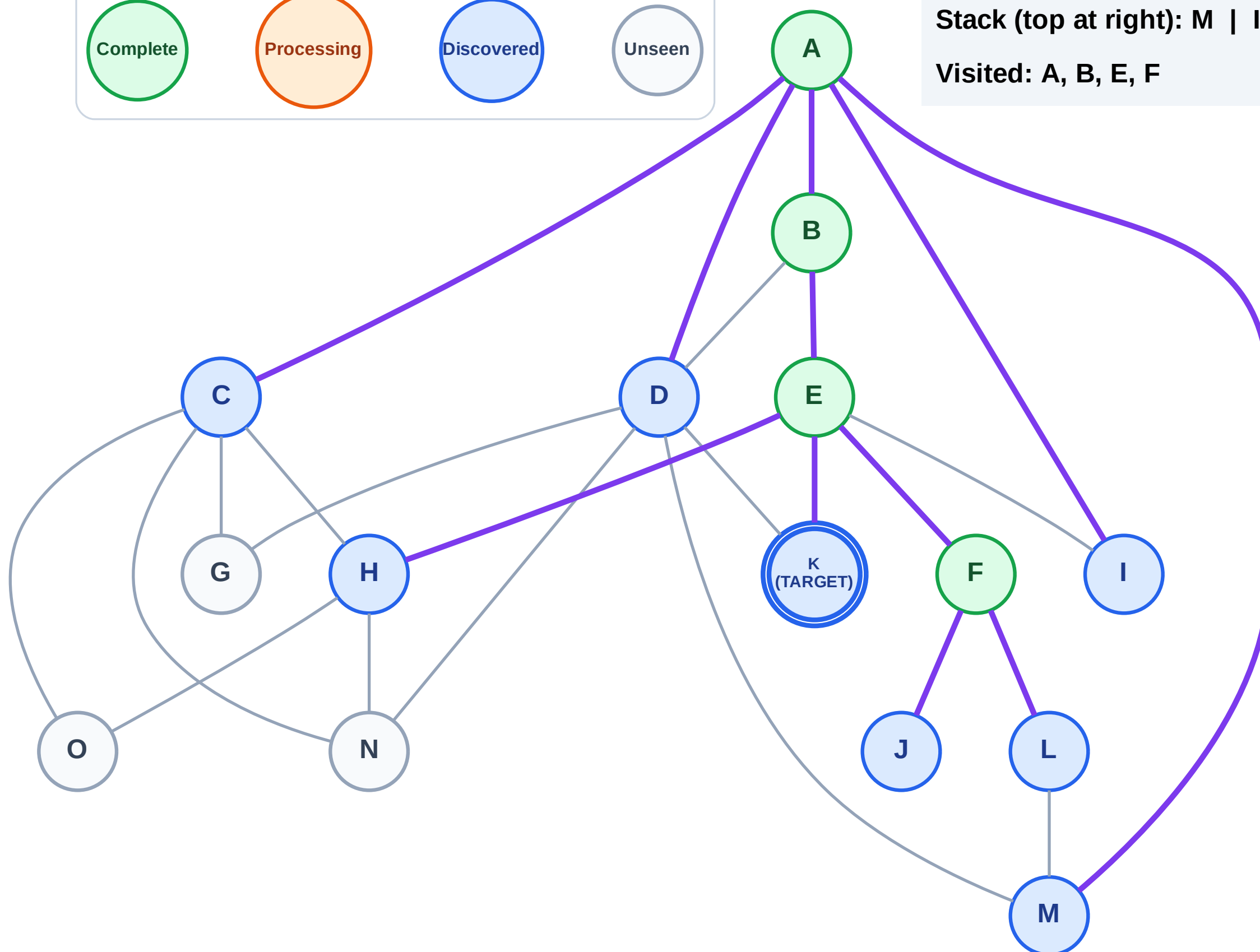
Step 9: Discover L, J from F

Legend



Stack (top at right): M | I | D | C | K | H | L | J

Visited: A, B, E, F



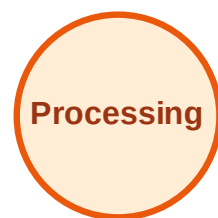
DFS Traversal

Step 10: Process node J

Legend



Complete



Processing



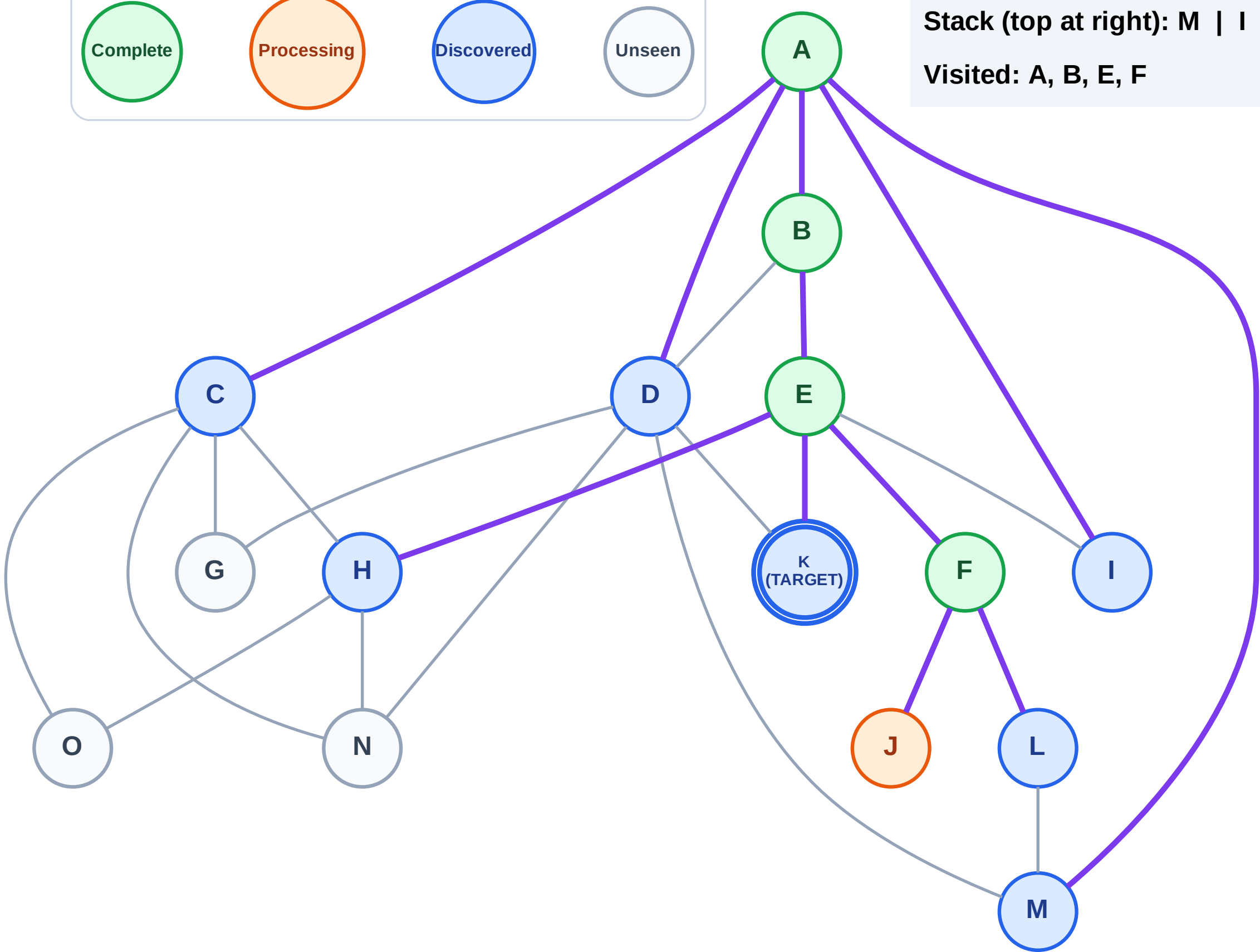
Discovered



Unseen

Stack (top at right): M | I | D | C | K | H | L

Visited: A, B, E, F



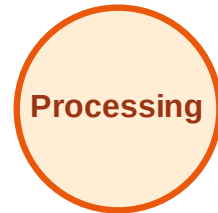
DFS Traversal

Step 11: No new neighbors from J

Legend



Complete



Processing



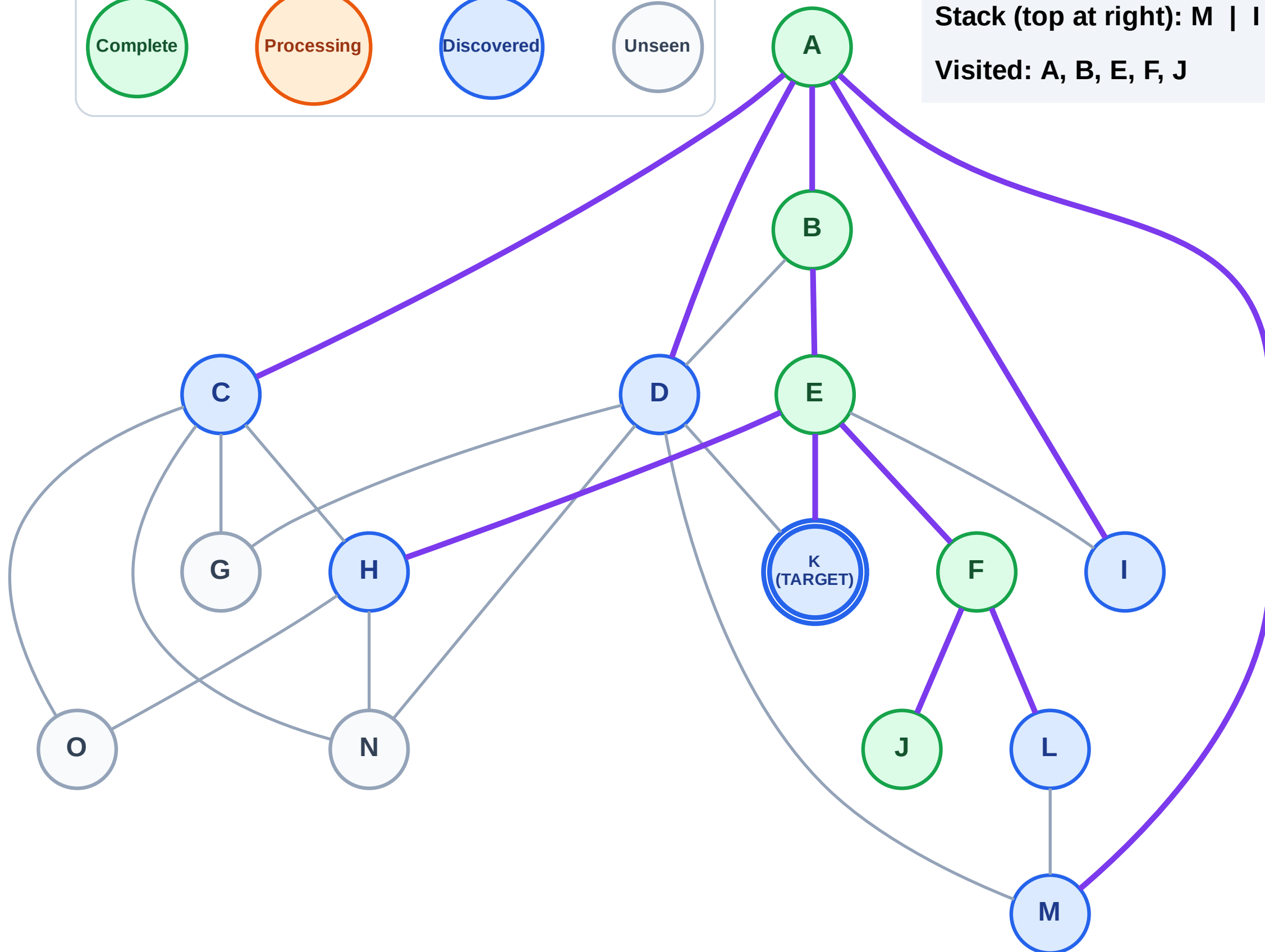
Discovered



Unseen

Stack (top at right): M | I | D | C | K | H | L

Visited: A, B, E, F, J



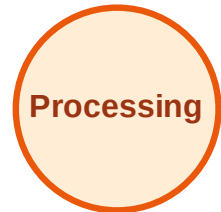
DFS Traversal

Step 12: Process node L

Legend



Complete



Processing



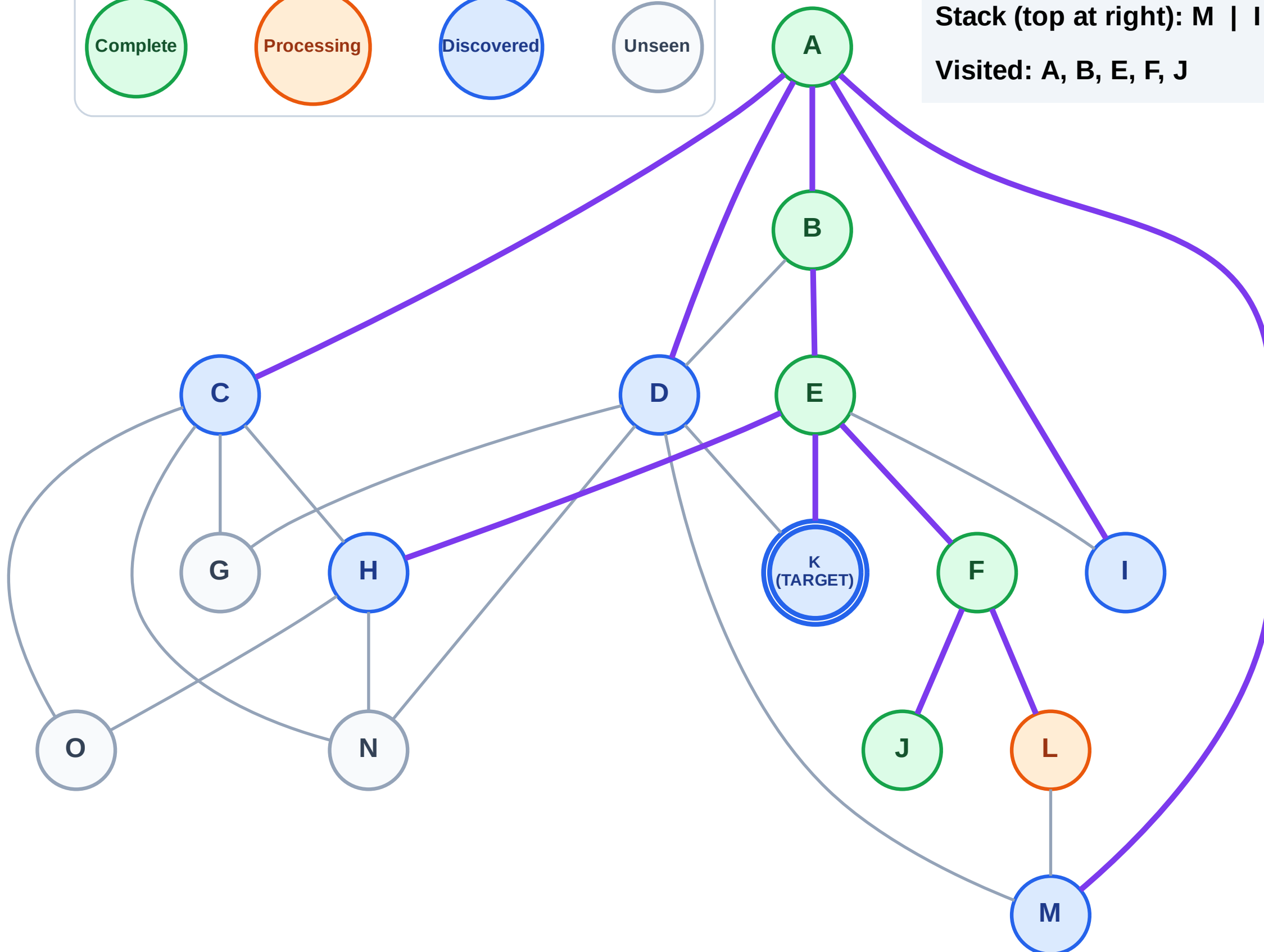
Discovered



Unseen

Stack (top at right): M | I | D | C | K | H

Visited: A, B, E, F, J



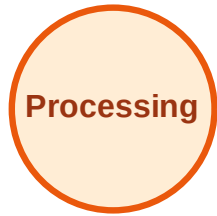
DFS Traversal

Step 13: No new neighbors from L

Legend



Complete



Processing



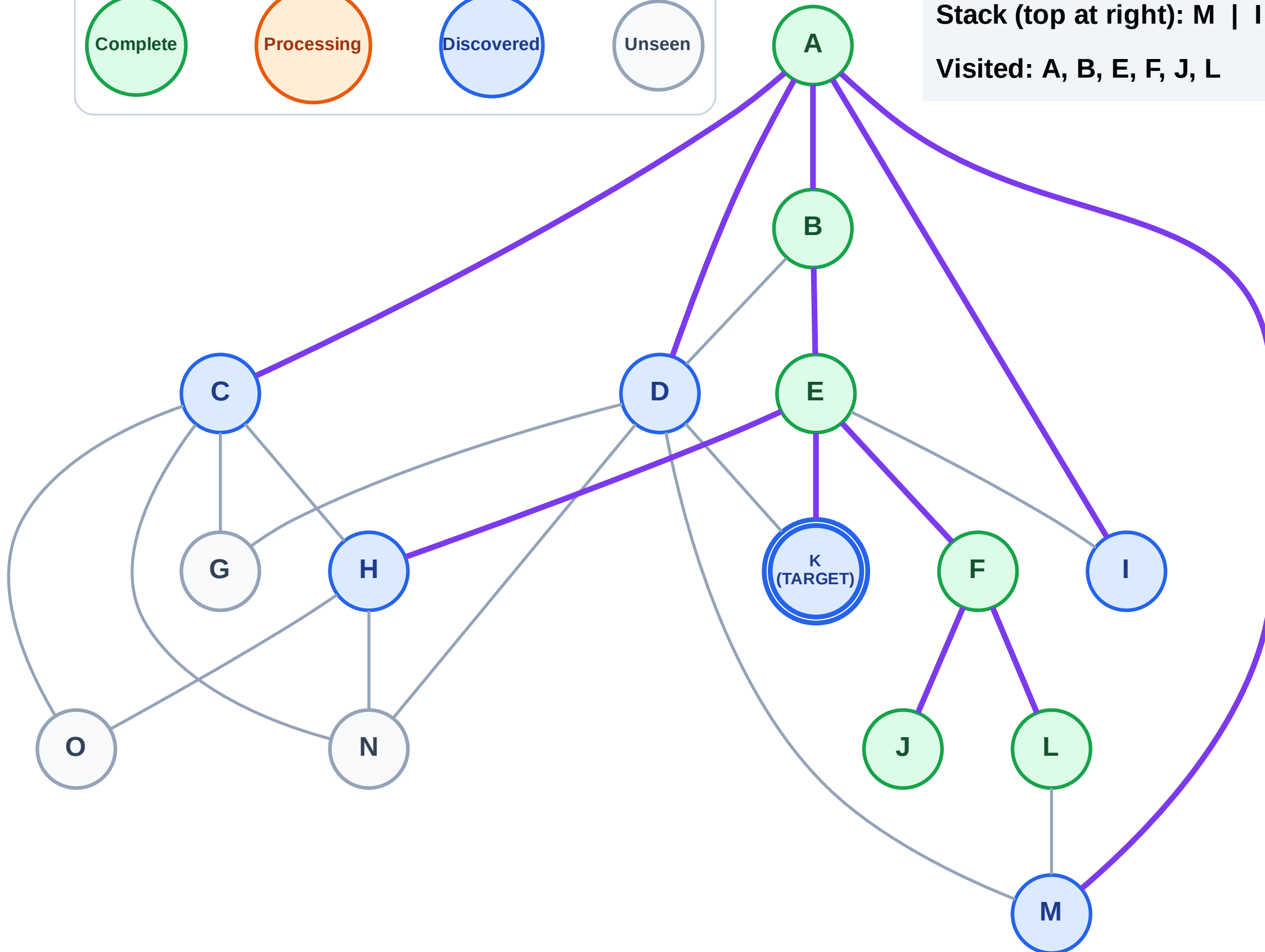
Discovered



Unseen

Stack (top at right): M | I | D | C | K | H

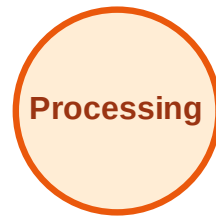
Visited: A, B, E, F, J, L



DFS Traversal

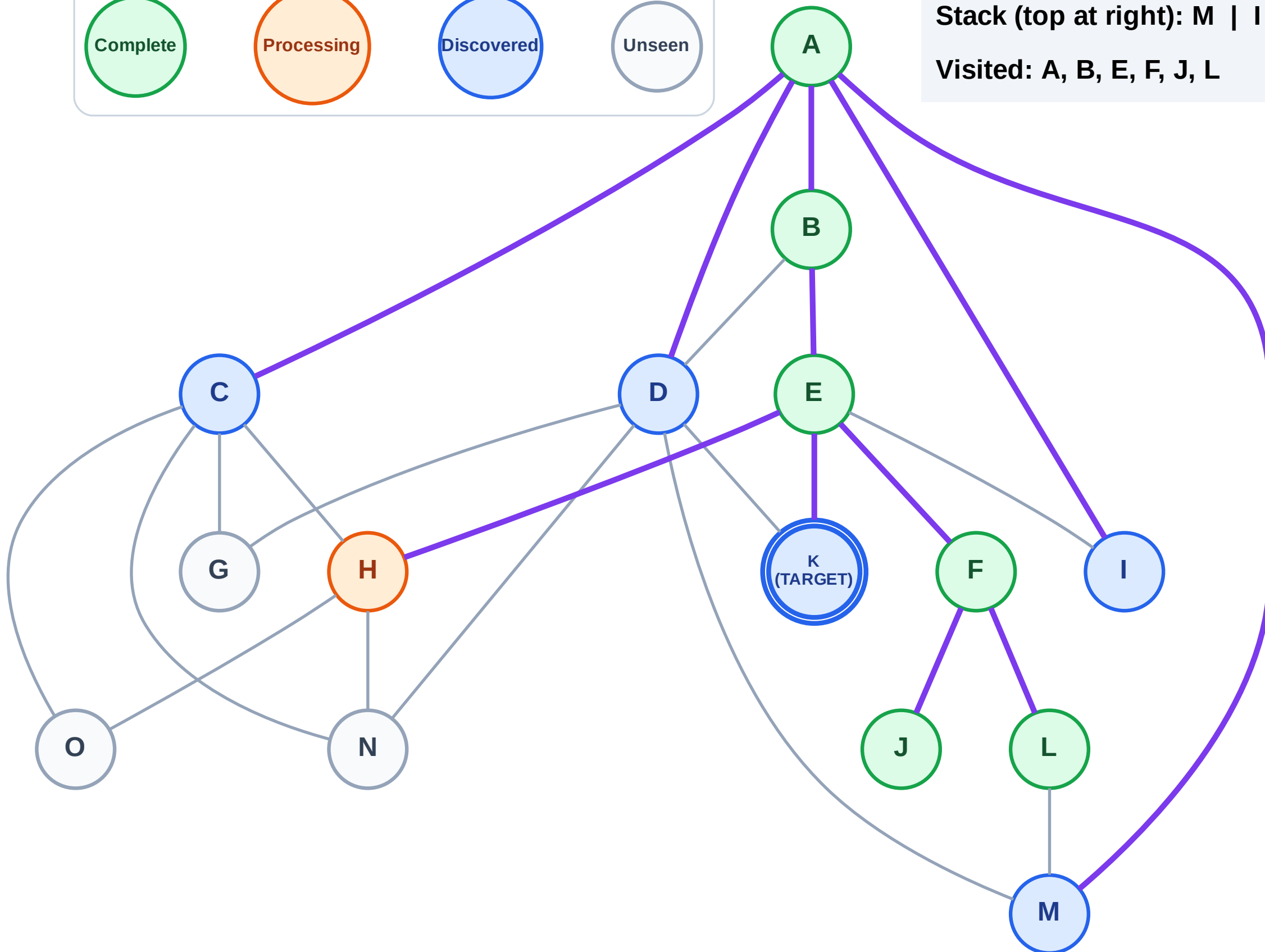
Step 14: Process node H

Legend



Stack (top at right): M | I | D | C | K

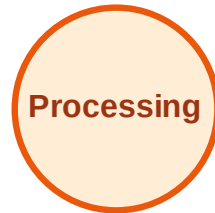
Visited: A, B, E, F, J, L



DFS Traversal

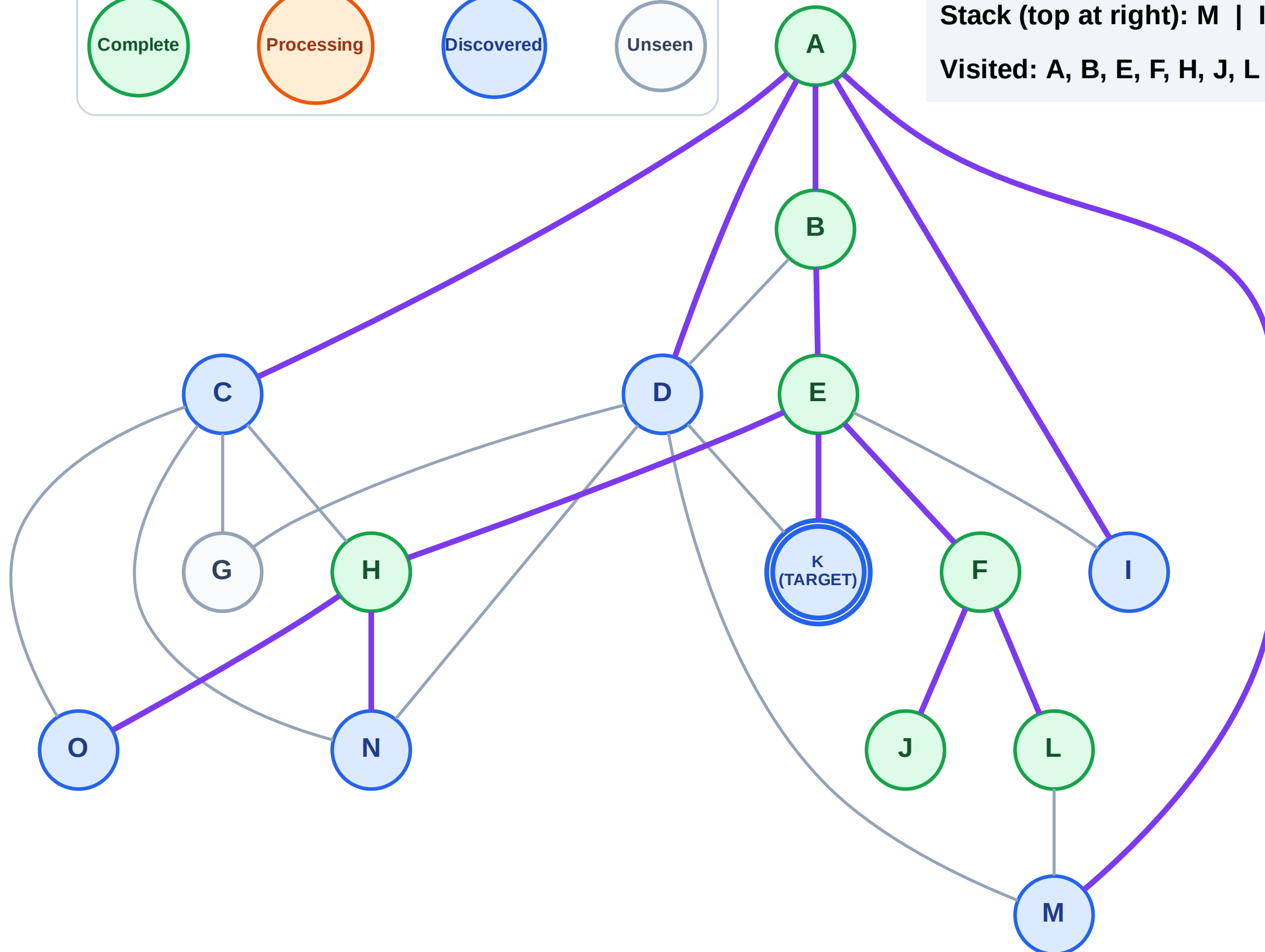
Step 15: Discover O, N from H

Legend



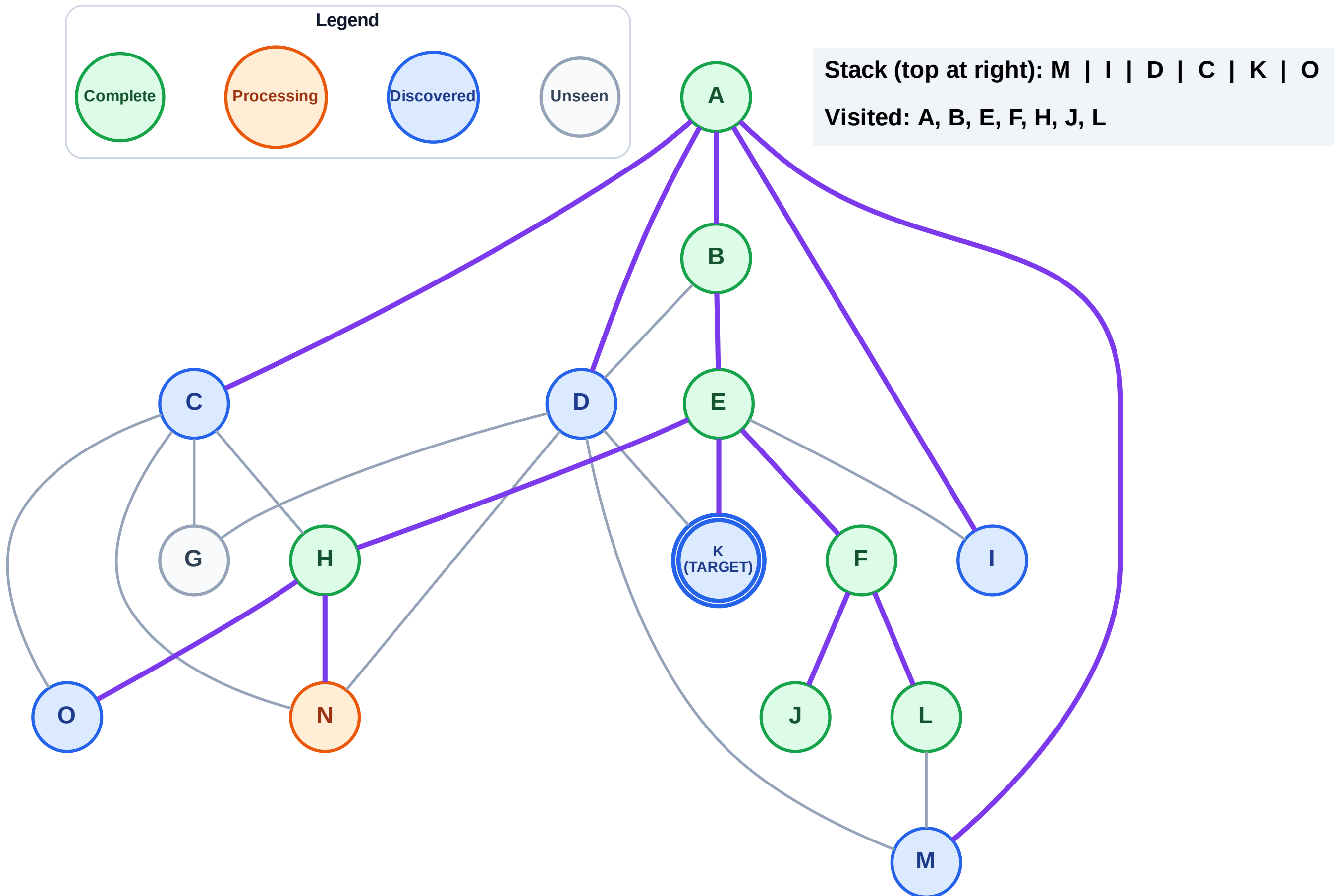
Stack (top at right): M | I | D | C | K | O | N

Visited: A, B, E, F, H, J, L



Step 16: Process node N

Step 16: Process node N



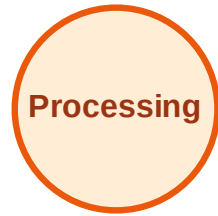
DFS Traversal

Step 17: No new neighbors from N

Legend



Complete



Processing



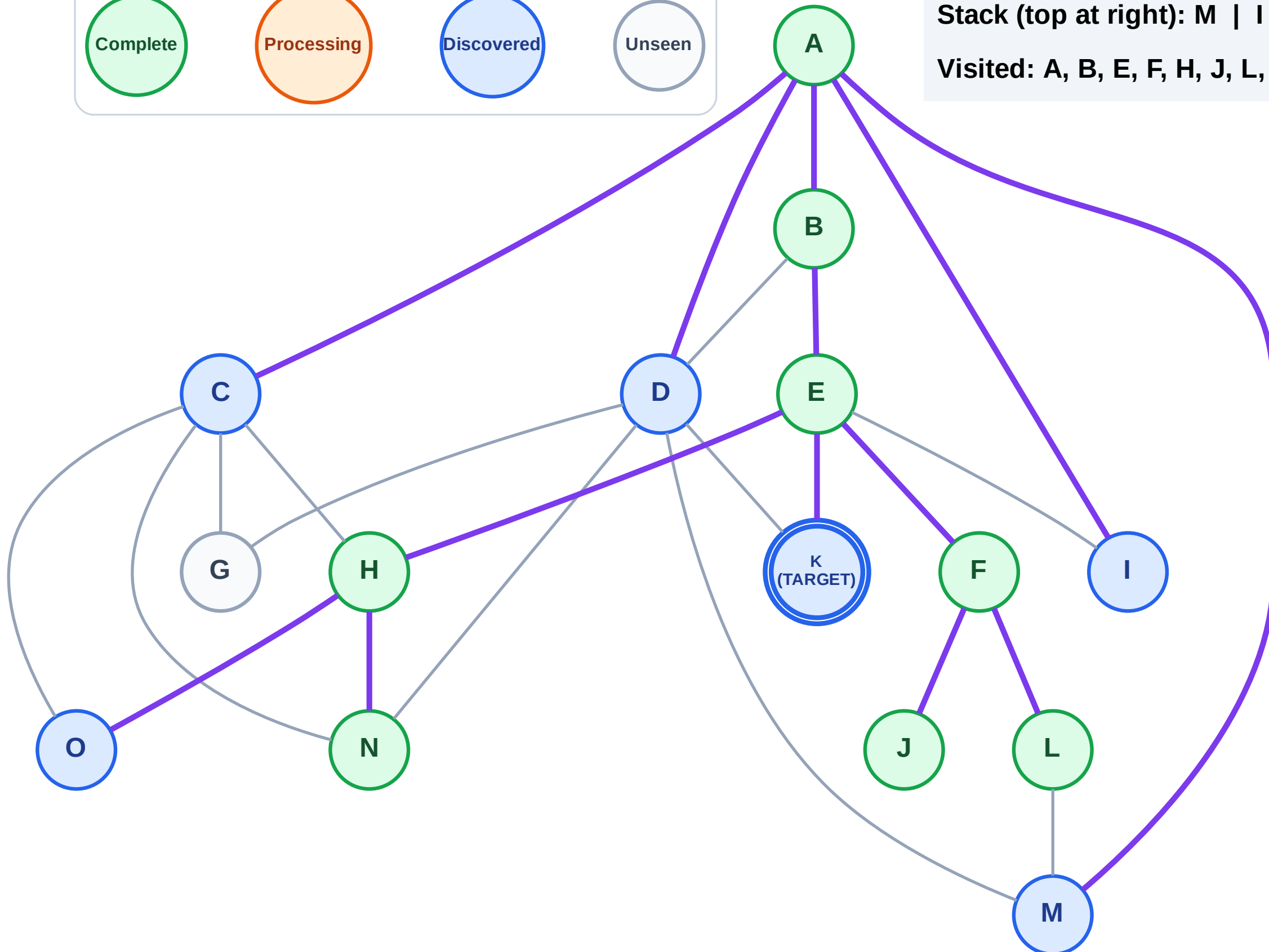
Discovered



Unseen

Stack (top at right): M | I | D | C | K | O

Visited: A, B, E, F, H, J, L, N



DFS Traversal

Step 18: Process node O

Legend

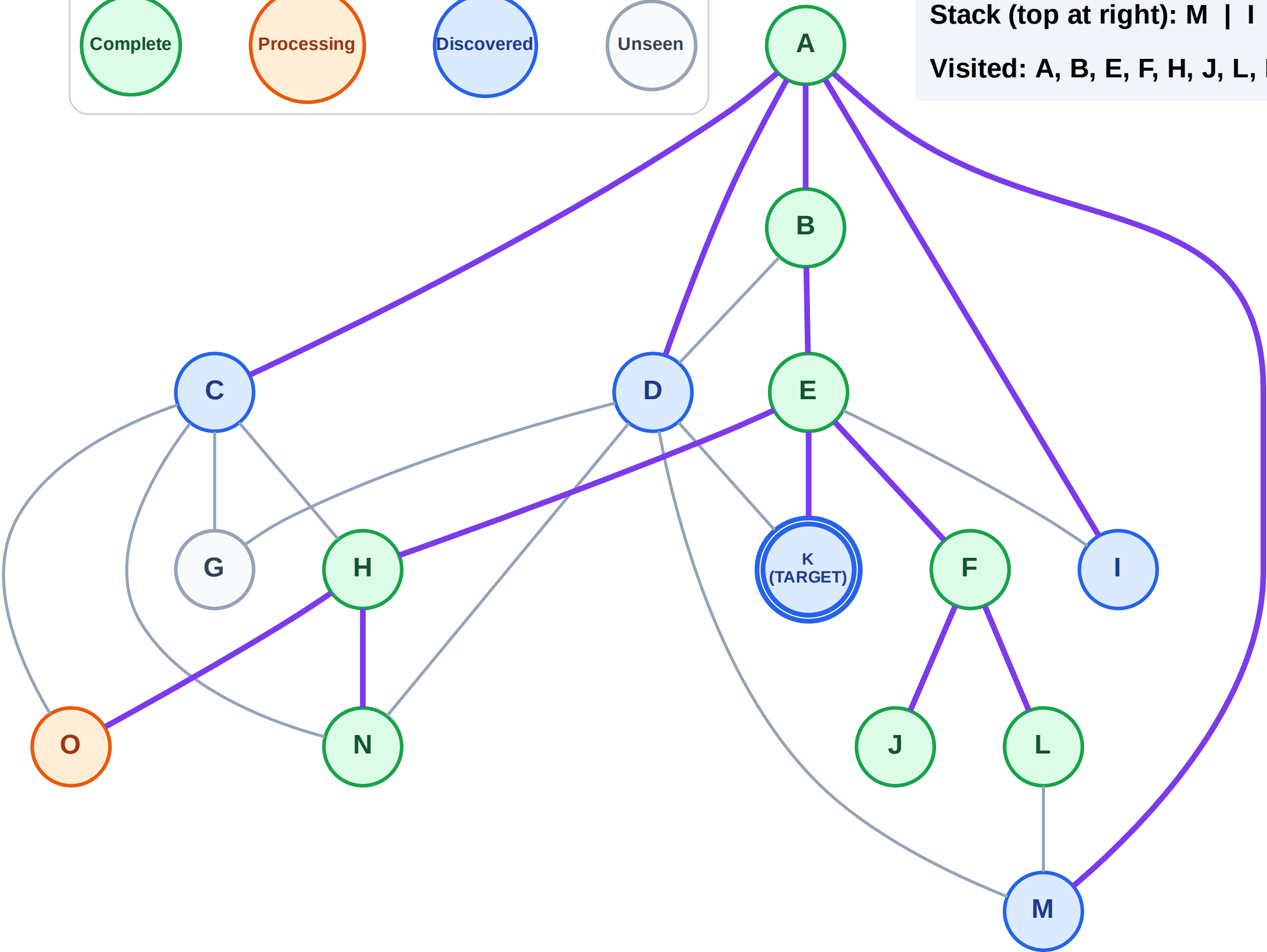
Complete

Processing

Discovered

Unseen

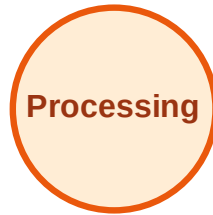
Stack (top at right): M | I | D | C | K
Visited: A, B, E, F, H, J, L, N



DFS Traversal

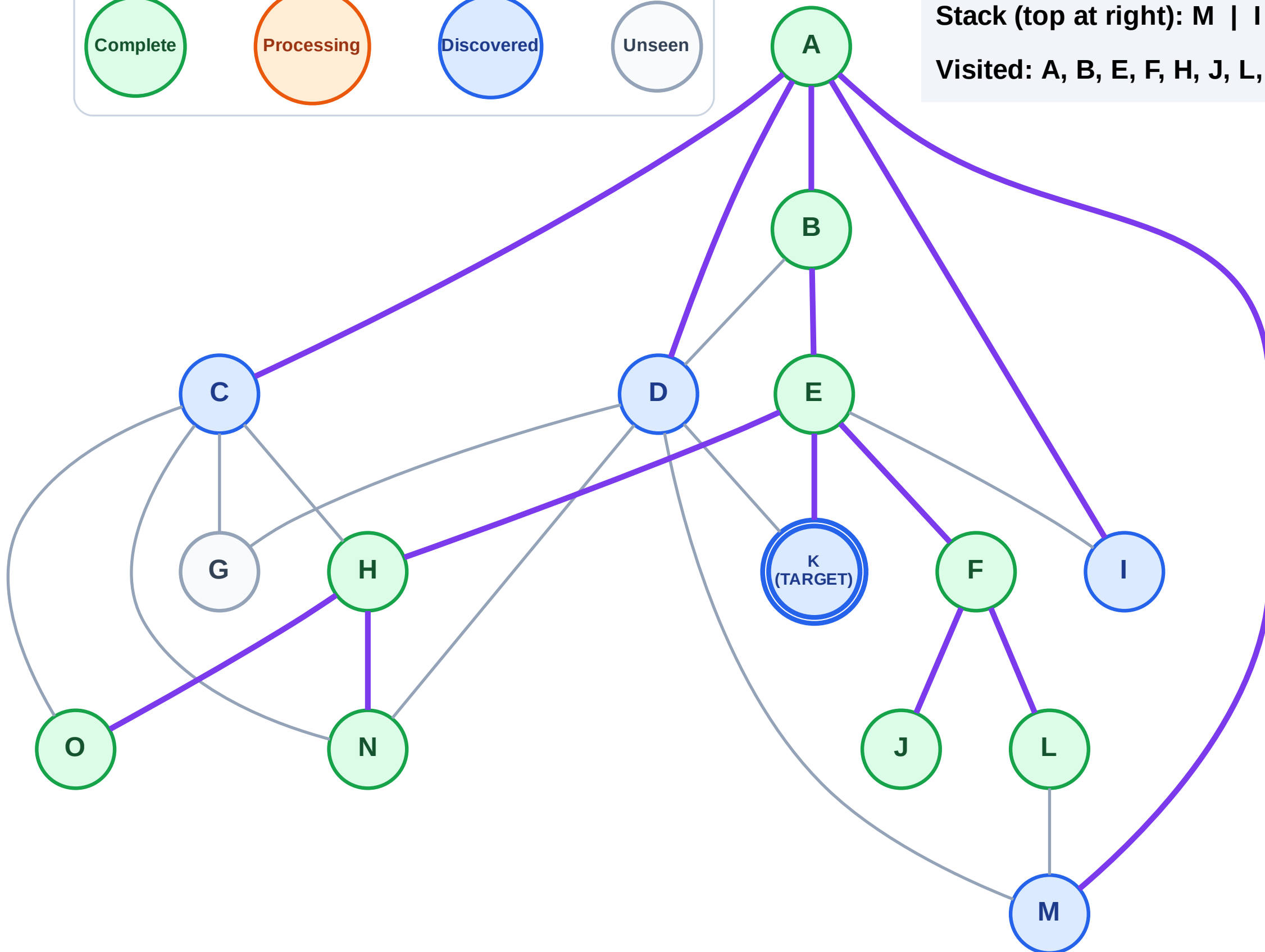
Step 19: No new neighbors from O

Legend



Stack (top at right): M | I | D | C | K

Visited: A, B, E, F, H, J, L, N, O



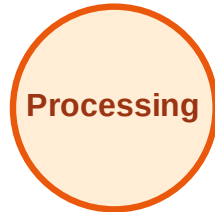
DFS Traversal

Step 20: Process node K

Legend



Complete



Processing



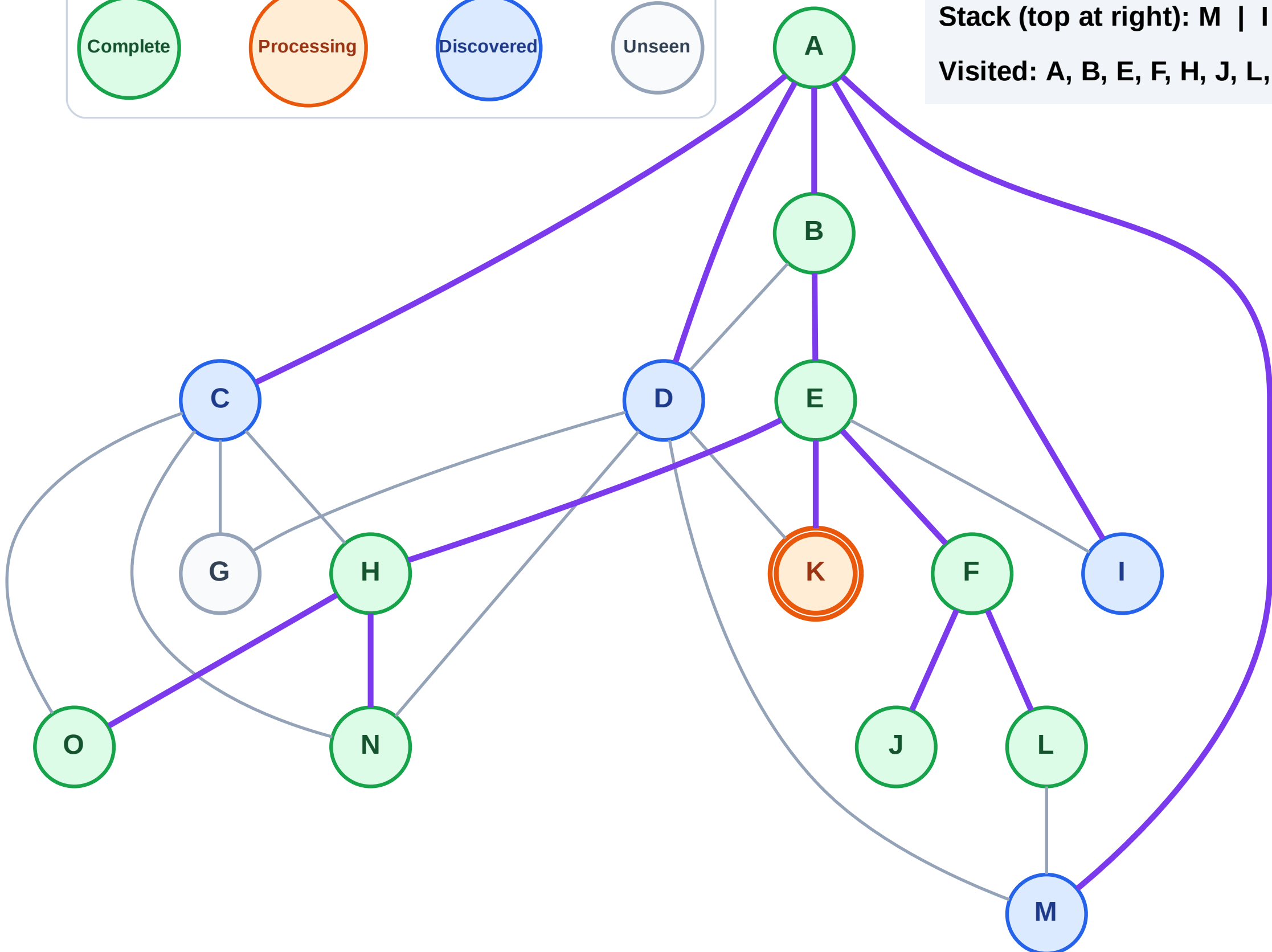
Discovered



Unseen

Stack (top at right): M | I | D | C

Visited: A, B, E, F, H, J, L, N, O



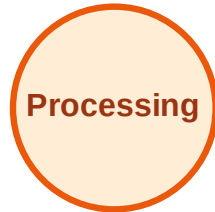
DFS Traversal

Step 21: TARGET NODE K REACHED! Path (3 hops): A → B → E → K

Legend



Complete



Processing



Discovered



Unseen

Stack (top at right): M | I | D | C

Visited: A, B, E, F, H, J, K, L, N, O

